

What is a Proof? ...What is Math?

A "fun" introduction to the maths you didn't know you loved.

Dr. Jim Ashe

IT College
Mathematics, Computer Science, Data Analytics, **Capstone**





Colbert on the Fields Medal.

Colbert Presents Talk on Union's Fields Medal Recipients

Cory Colbert, assistant professor of mathematics at W&L, will present on this year's International Mathematical Union's Fields Medal on Wednesday, Nov. 16.

By W&L News Office
November 15, 2022



Cory Colbert

What is this stuff?

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- Proof?

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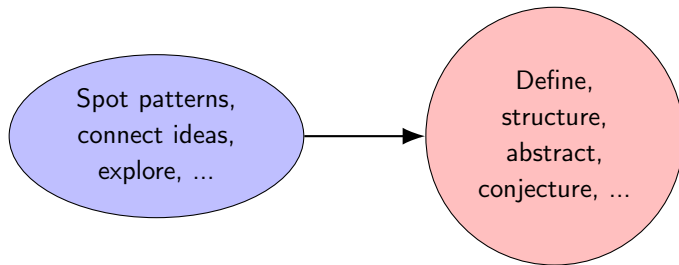
- Proof?
- Conjecture?

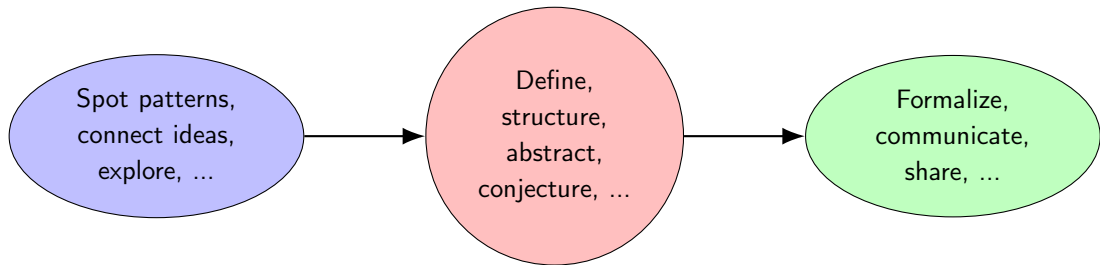
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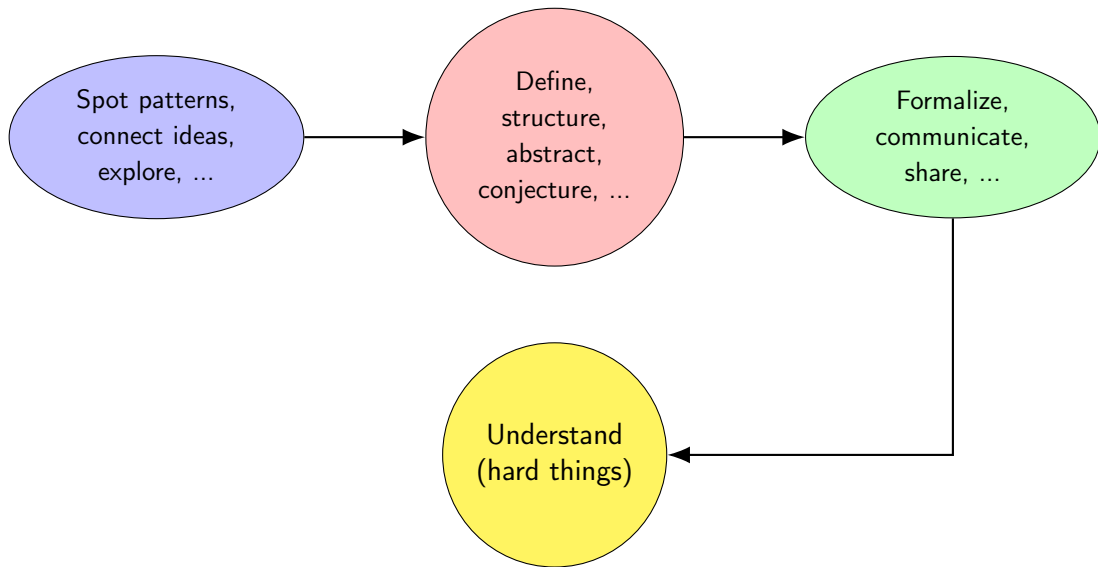
- Proof?
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- Weird Stuff with a bunny?

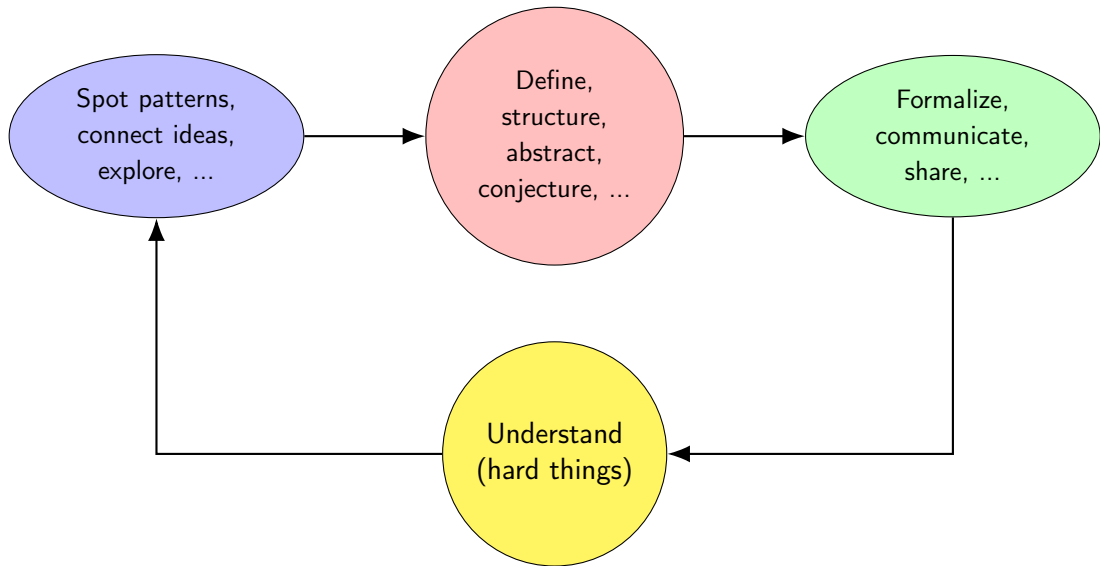


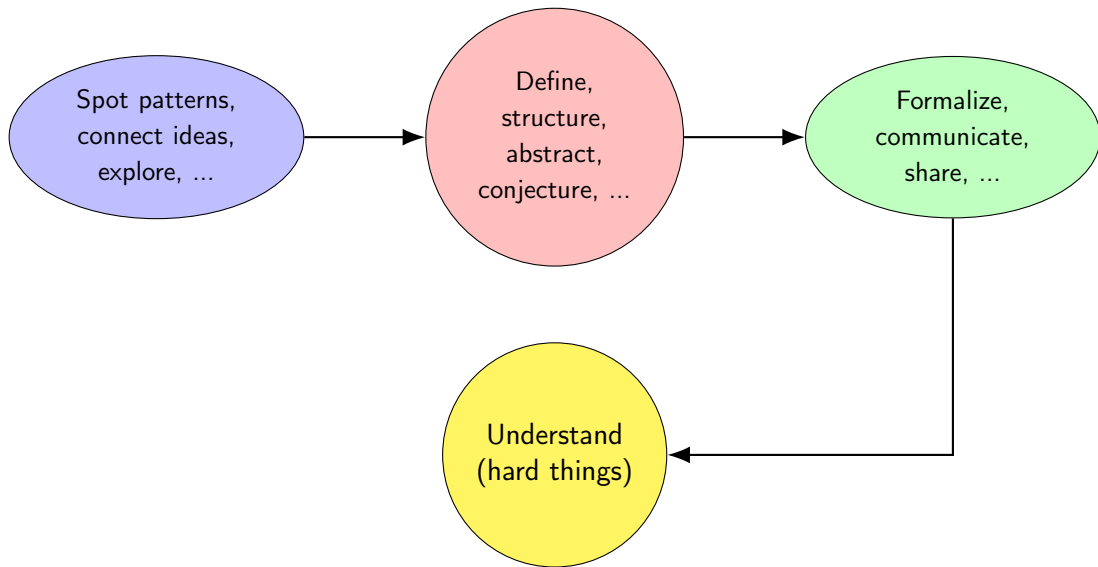
Spot patterns,
connect ideas,
explore, ...



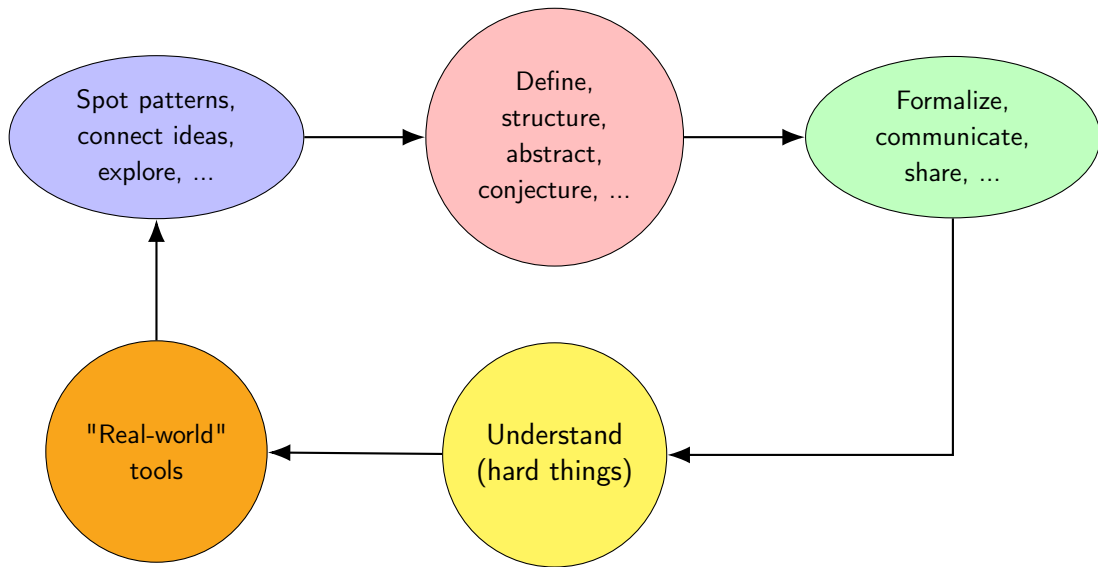




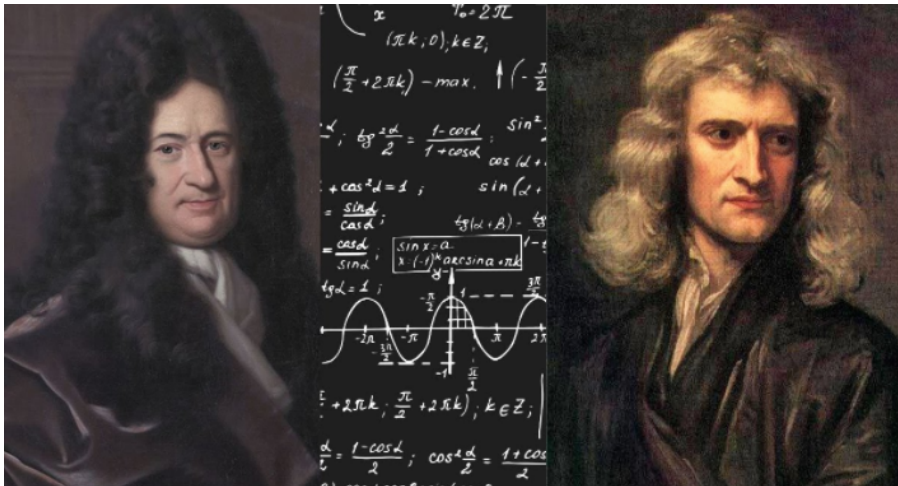




The language of this process is "proof," and logic is its grammar.



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"I have never done anything 'useful'. No discovery of mine has made, or is likely to make, directly or indirectly, for good or ill, the least difference to the amenity of the world."

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"Vector analysis is interesting, but it will never have any practical application."

-William Thomson (Lord Kelvin) 1824-1907

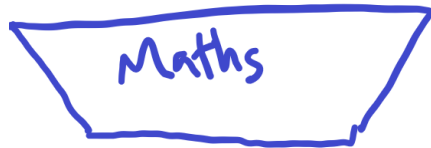
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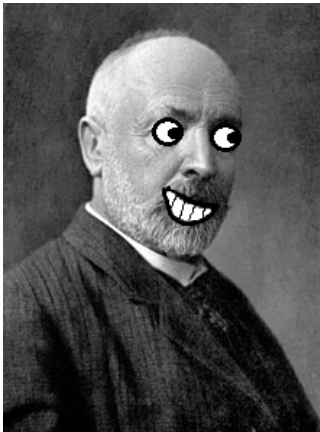
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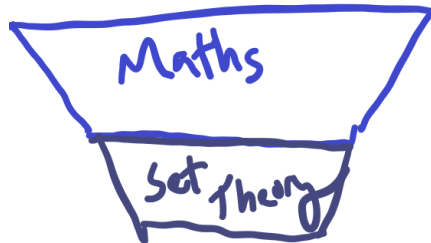
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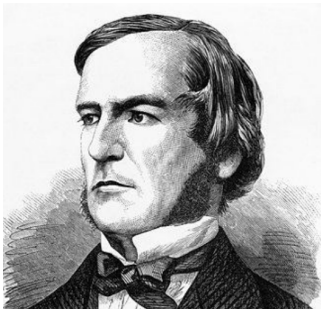
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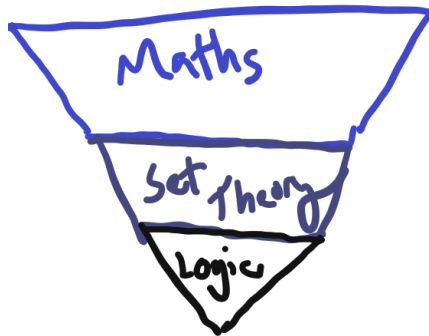
Georg Cantor (1845-1918)



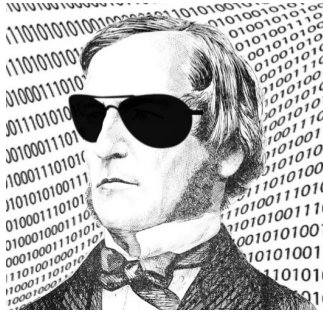
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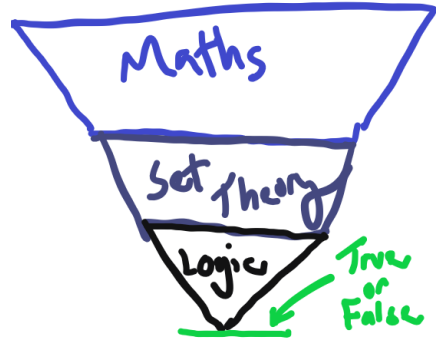
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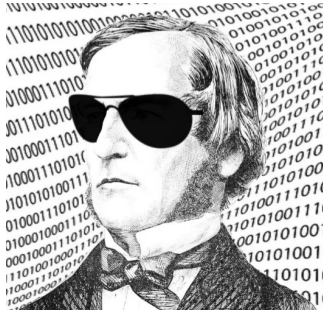
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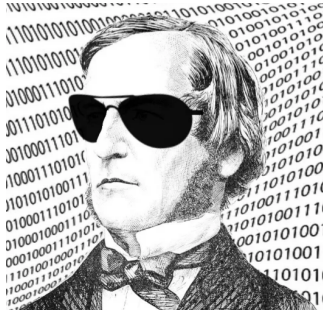
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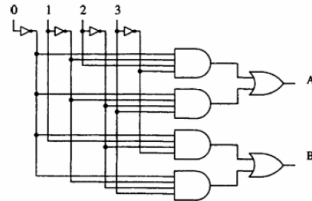
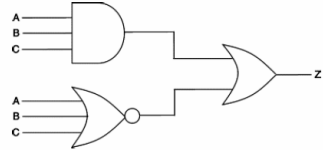
Source registers				
R1	0100 0110	1010 0001	1111 0001	1011 0111
R2	1111 1111	1111 1111	0000 0000	0000 0000

Assembly code		Result			
AND R3, R1, R2	R3	0100 0110	1010 0001	0000 0000	0000 0000
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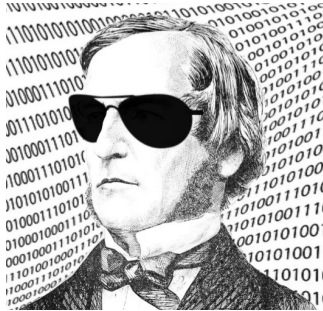
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A way to communicate absolute truths, through logic.

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```
def valid_words_mask(sentence):
    result = []
    temp = [w for w in sentence.split(' ')]
    count = 0
    for w in temp:
        chk = True
        if any(l.isdigit() for l in w):
            chk = False
        if w.isalpha() is False:
            p = [l for l in list(w) if l.isalnum() is False]
            if len(p) > 1:
                chk = False
            elif len(p) > 0 and w[len(w) - 1] != p[0]:
                chk = False
        h = [l for l in list(w) if l == '-']
        if len(h) > 0:
            if len(h) > 1:
                chk = False
            elif w[0] == '-' or w[len(w) - 1] == '-':
                chk = False
    return result
```

You've all done this.

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$$x = 5 \text{ or } -3$$

- Direct proof.

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- Proof by contradiction.

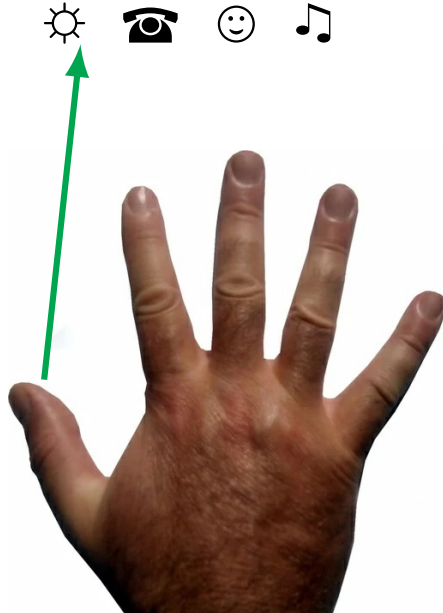
- Direct proof.
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- Proof by induction.



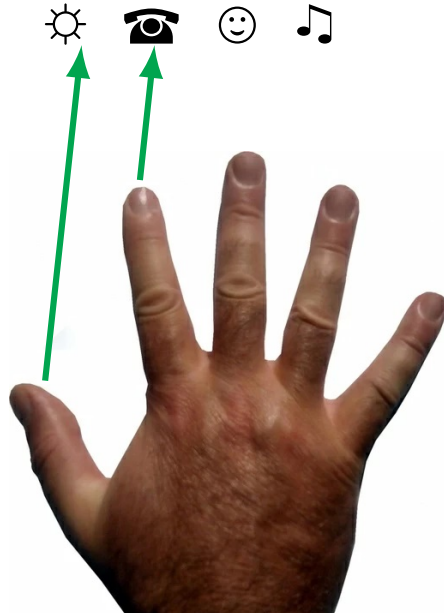
More fingers
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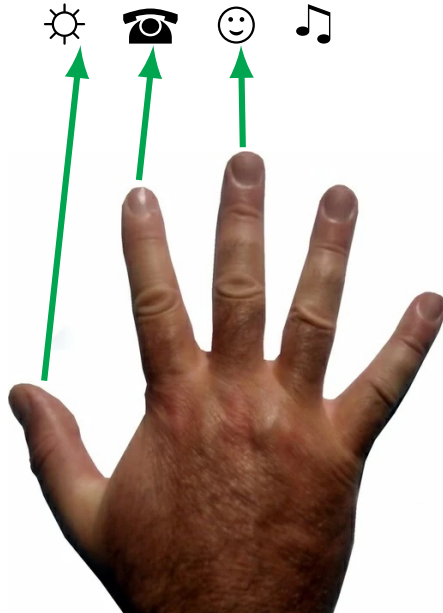
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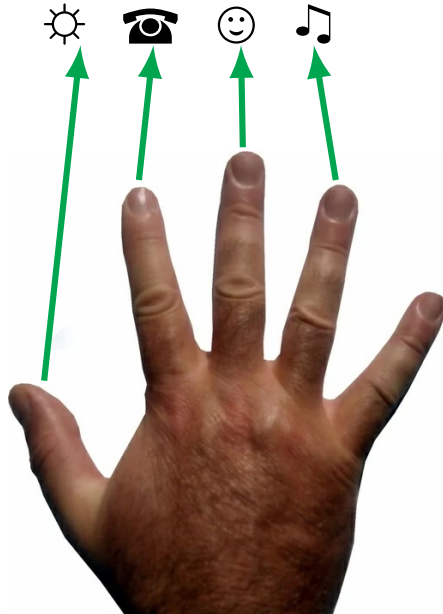
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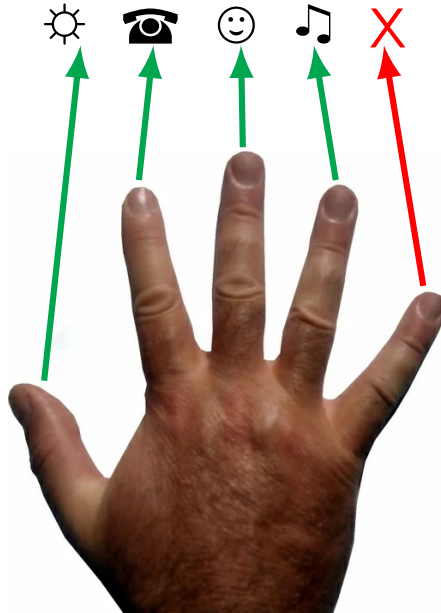
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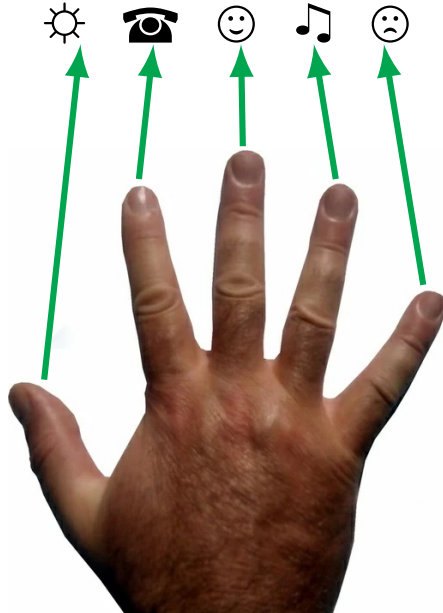
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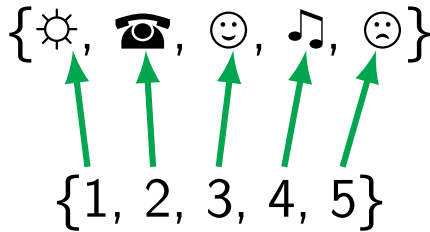
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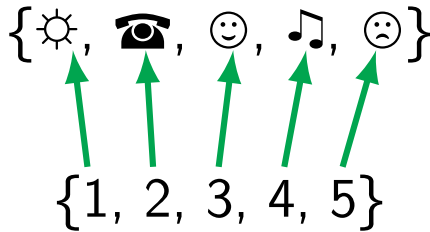


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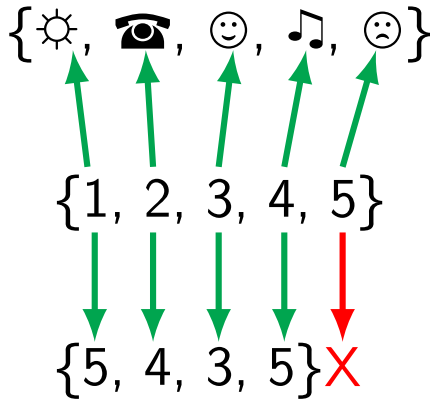


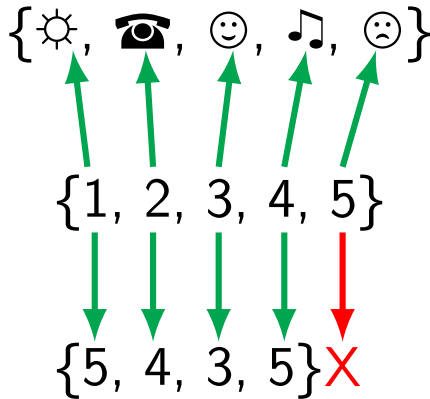




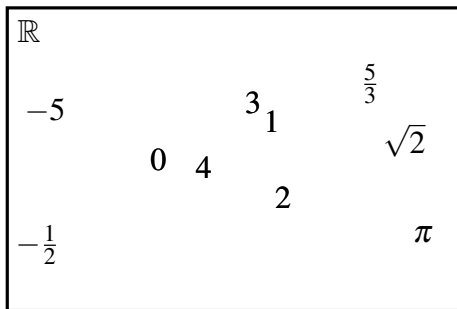


{5, 4, 3, 5}

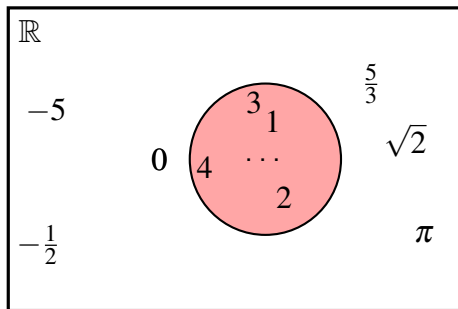




"The essence of math is not to make simple things complicated,
but to make *complicated things simple*."



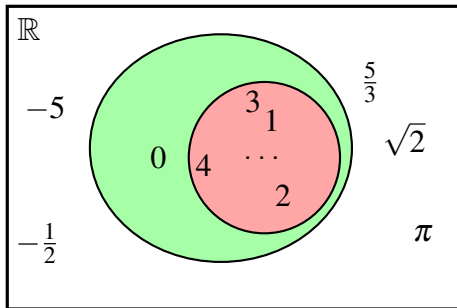
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Are these two sets the same?

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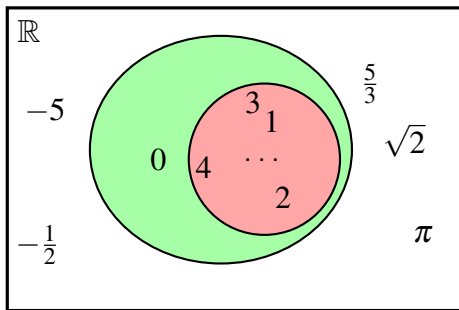


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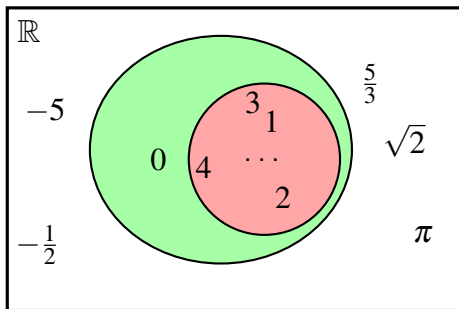
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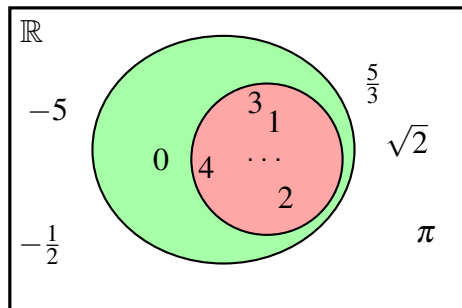
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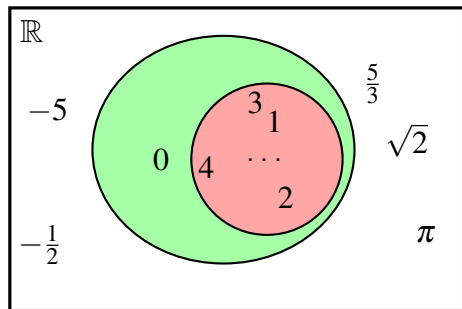
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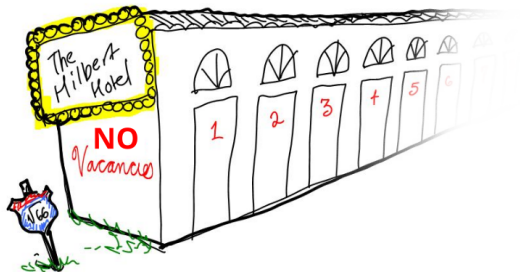
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The Hilbert Hotel Problem



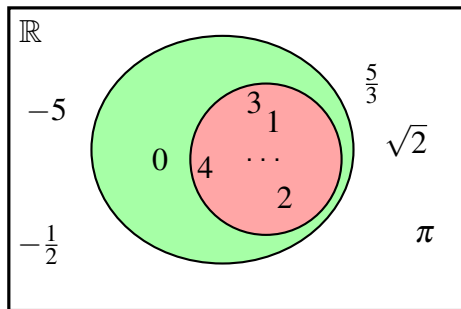
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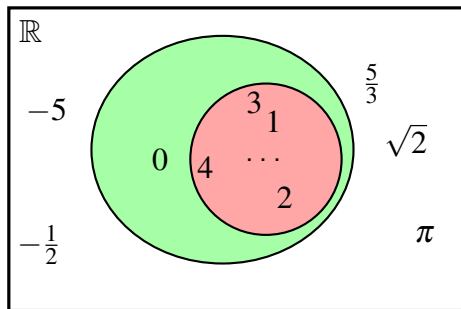
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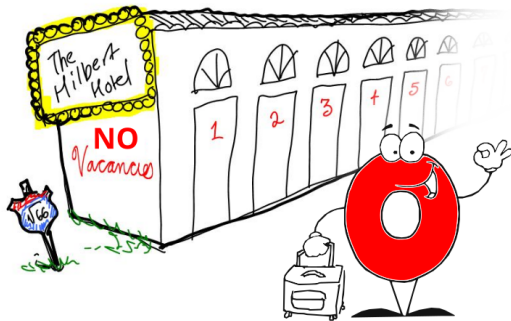
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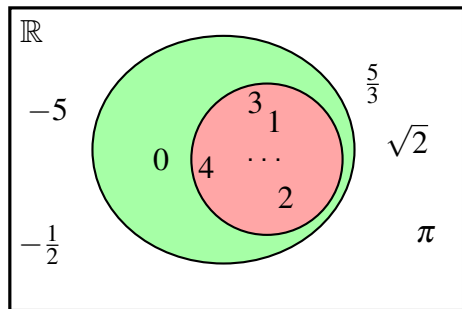
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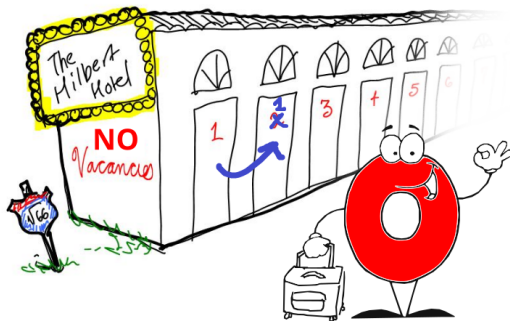
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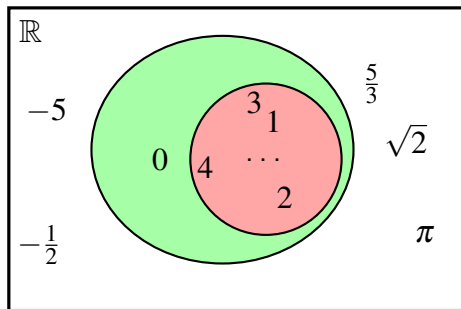
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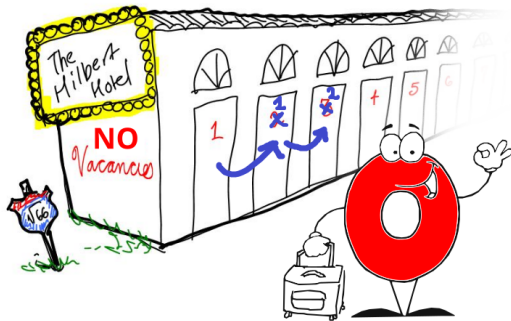
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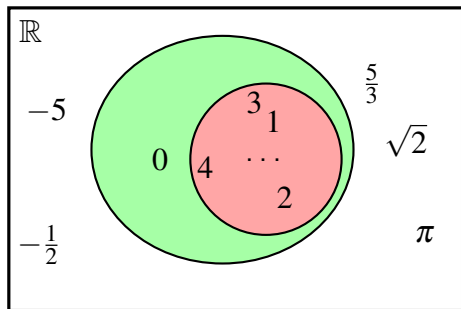
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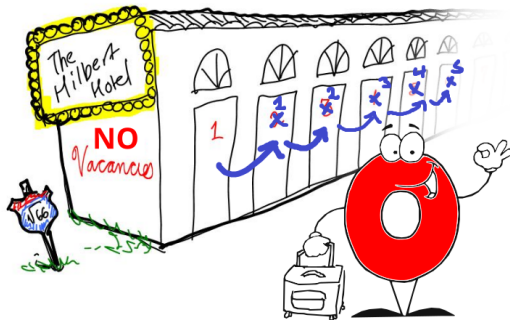
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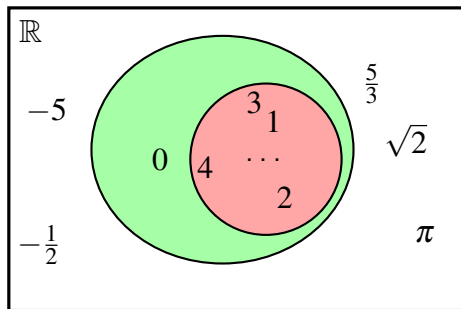
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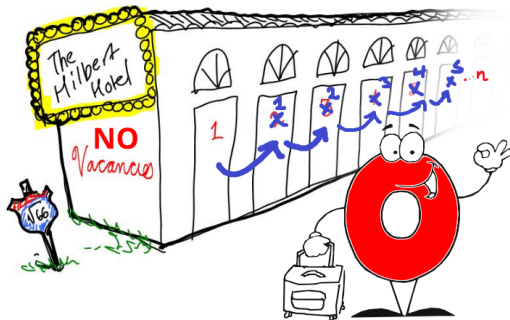
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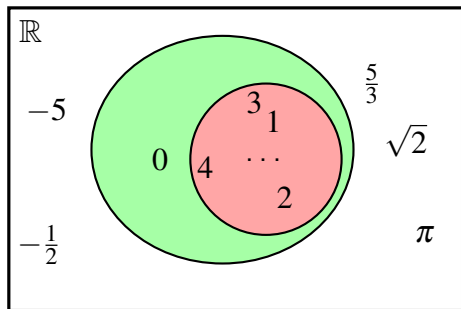
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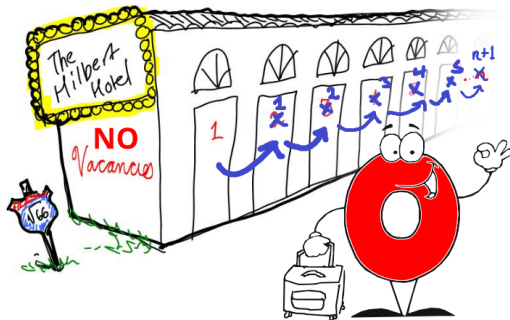
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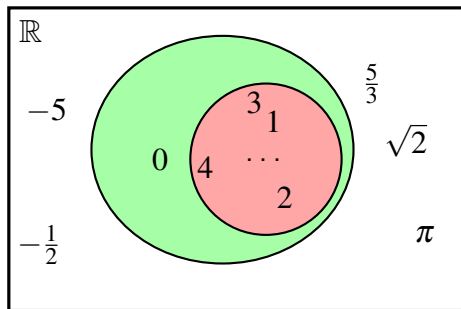
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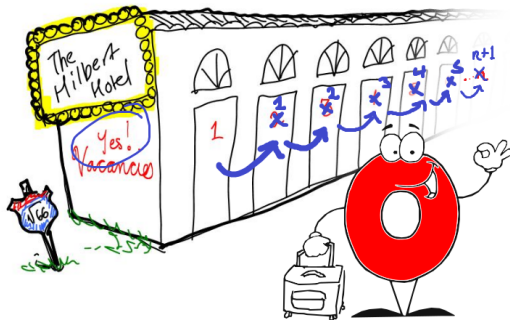
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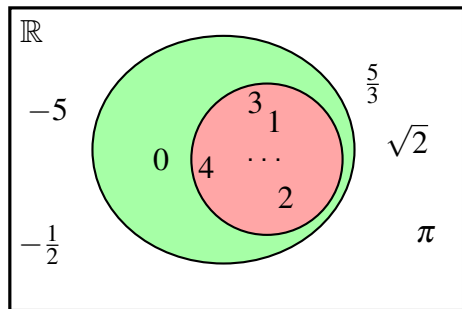
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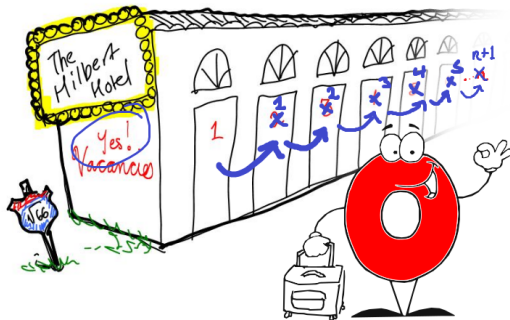
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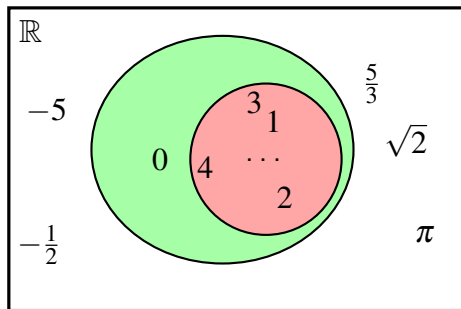
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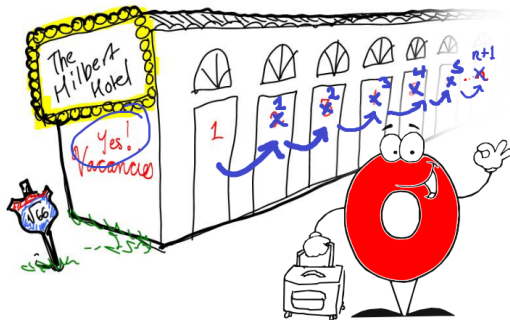
Yes!

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

$$\mathbb{N}_0 = \{0, 1, 2, 3, 4, \dots\}$$



The Hilbert Hotel Problem



We call these sets *countable*

Are these two sets the same?

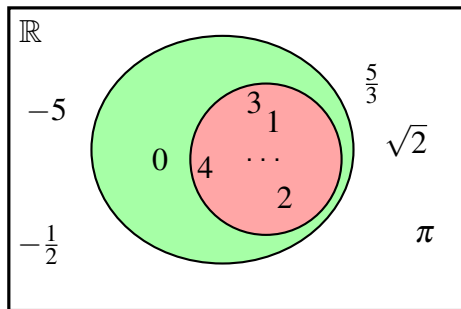
No.

Are these two sets the same *size*?

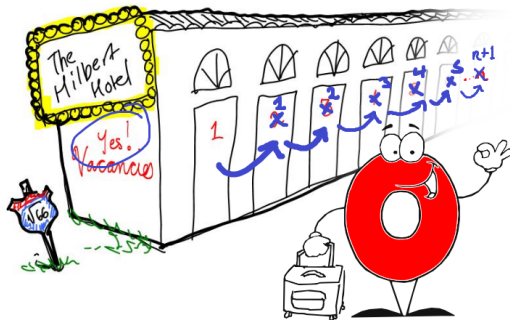
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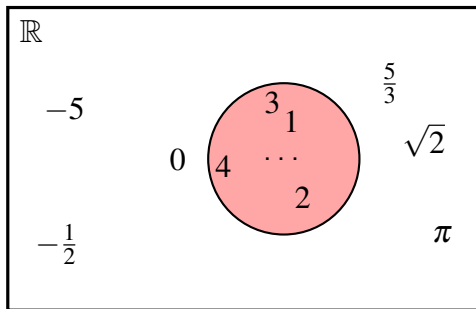
The Hilbert Hotel Problem



We call these sets *countable* ... because we can *count* them.

Are these two sets the same size?

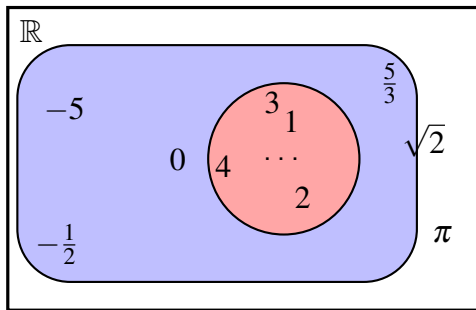
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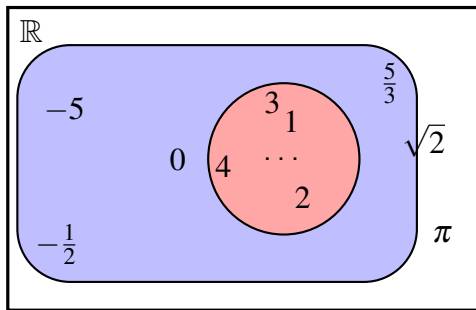
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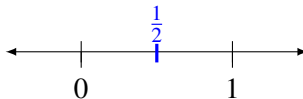
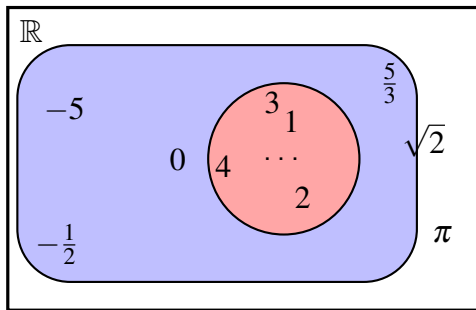
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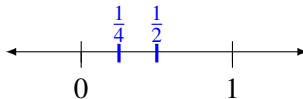
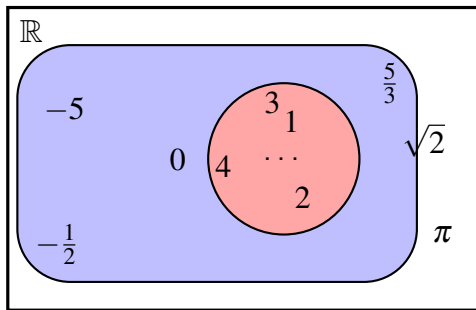
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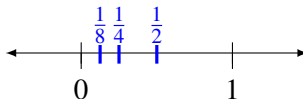
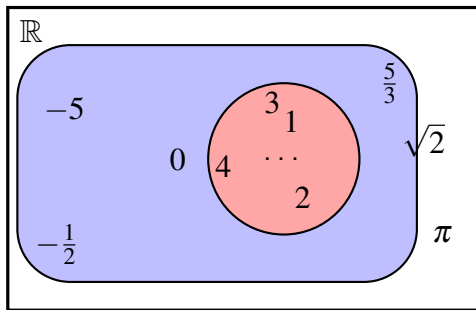
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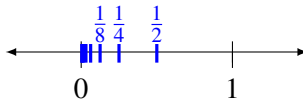
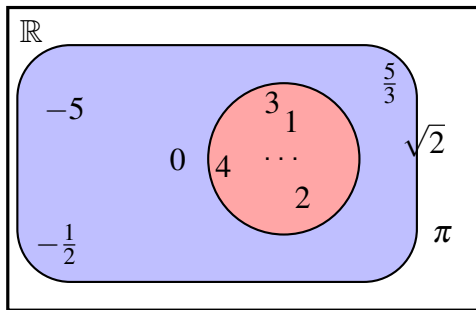
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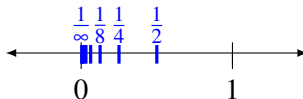
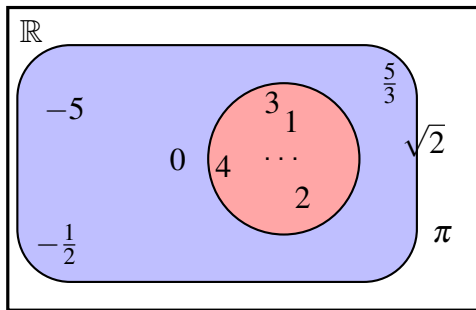
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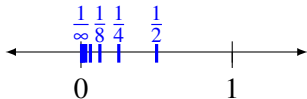
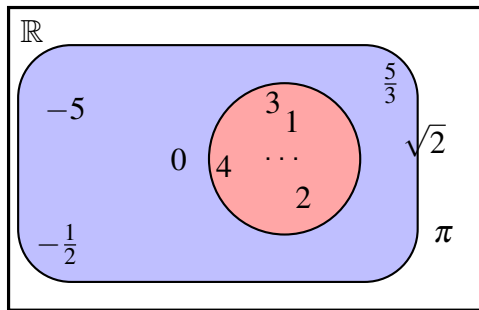
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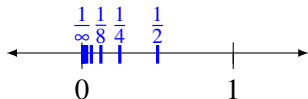
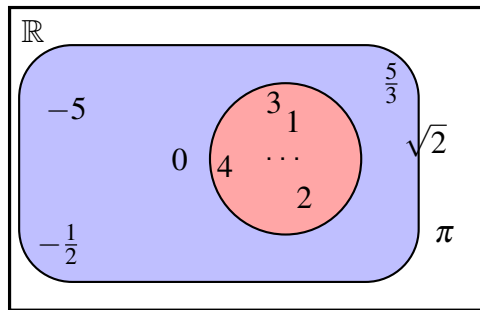
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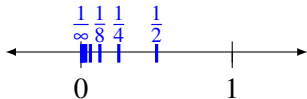
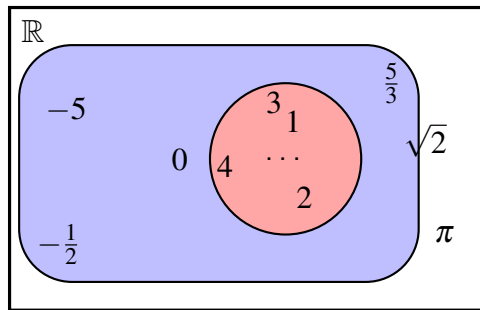


$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\dots$
$\frac{2}{1}$	$\frac{2}{2}$	$\frac{2}{3}$	$\frac{2}{4}$	$\frac{2}{5}$	$\frac{2}{6}$	$\dots$
$\frac{3}{1}$	$\frac{3}{2}$	$\frac{3}{3}$	$\frac{3}{4}$	$\frac{3}{5}$	$\frac{3}{6}$	$\dots$
$\frac{4}{1}$	$\frac{4}{2}$	$\frac{4}{3}$	$\frac{4}{4}$	$\frac{4}{5}$	$\frac{4}{6}$	$\dots$
$\frac{5}{1}$	$\frac{5}{2}$	$\frac{5}{3}$	$\frac{5}{4}$	$\frac{5}{5}$	$\frac{5}{6}$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

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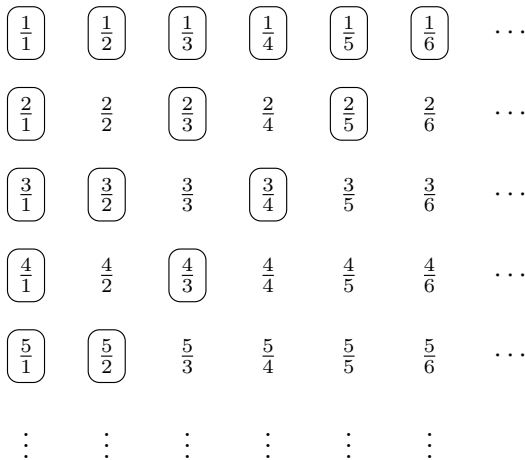
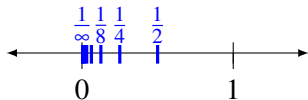
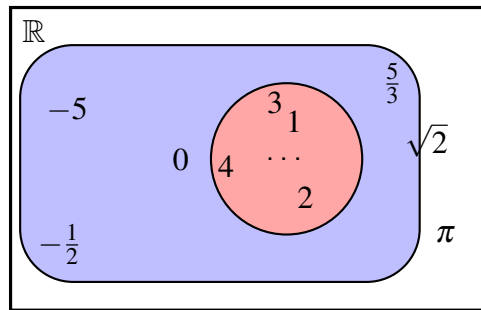


$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	...
$\frac{2}{1}$	$\frac{2}{2}$	$\frac{2}{3}$	$\frac{2}{4}$	$\frac{2}{5}$	$\frac{2}{6}$	...
$\frac{3}{1}$	$\frac{3}{2}$	$\frac{3}{3}$	$\frac{3}{4}$	$\frac{3}{5}$	$\frac{3}{6}$	...
$\frac{4}{1}$	$\frac{4}{2}$	$\frac{4}{3}$	$\frac{4}{4}$	$\frac{4}{5}$	$\frac{4}{6}$	...
$\frac{5}{1}$	$\frac{5}{2}$	$\frac{5}{3}$	$\frac{5}{4}$	$\frac{5}{5}$	$\frac{5}{6}$	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

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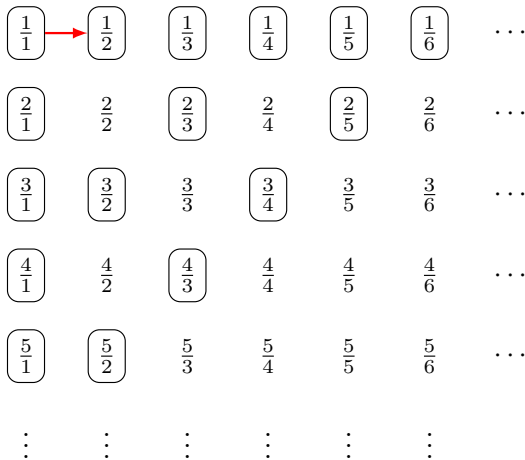
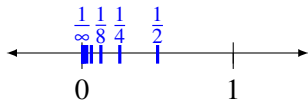
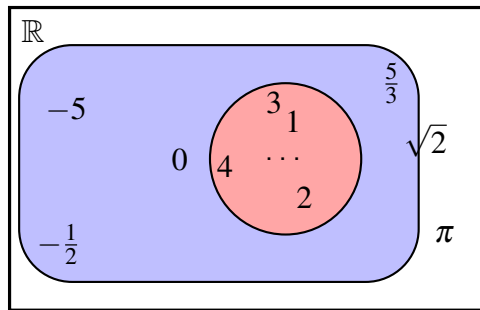
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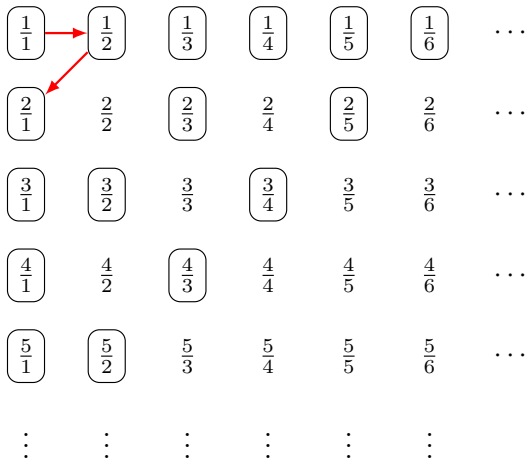
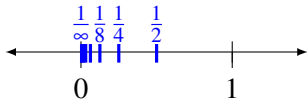
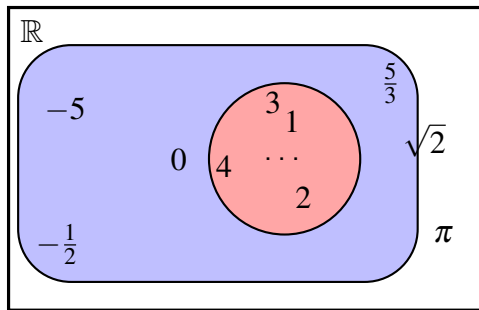
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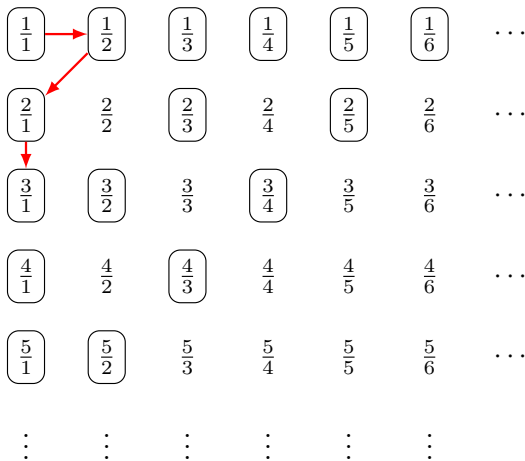
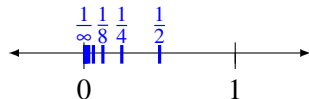
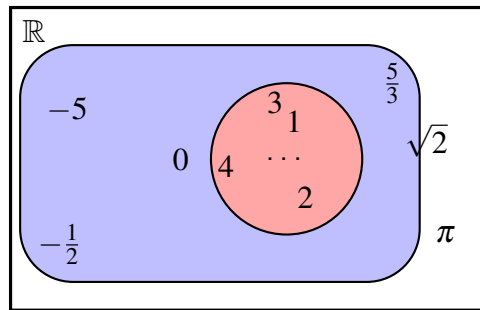
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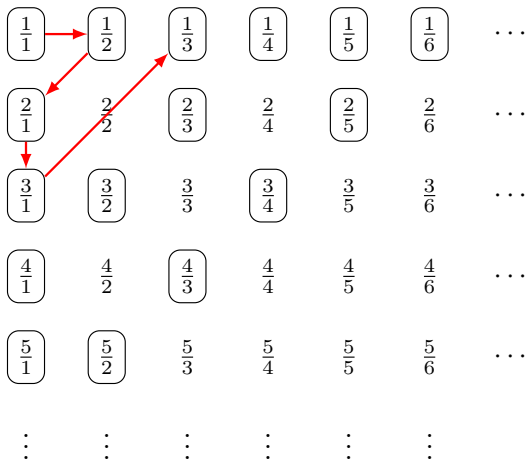
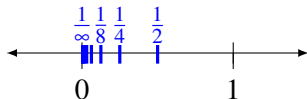
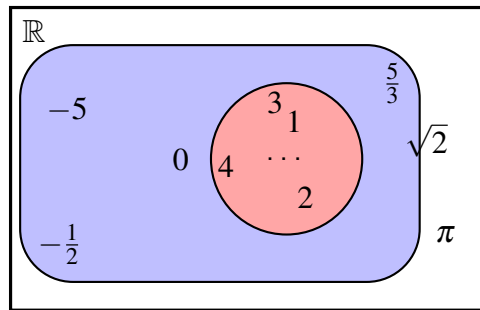
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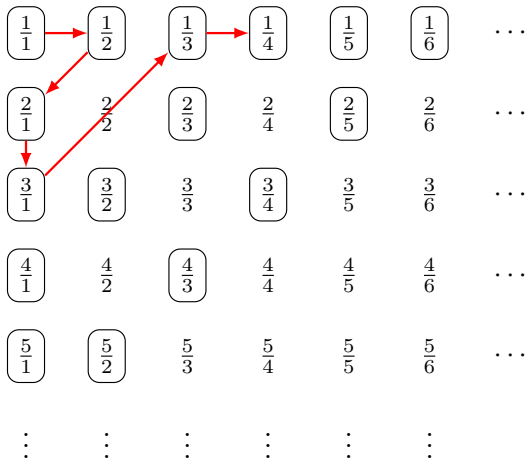
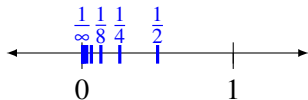
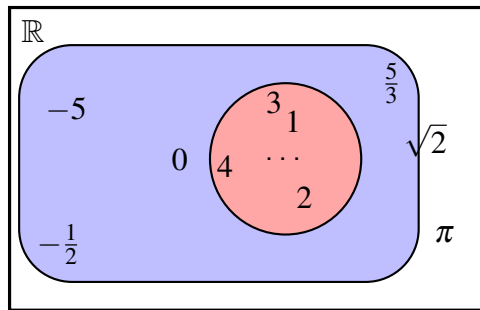
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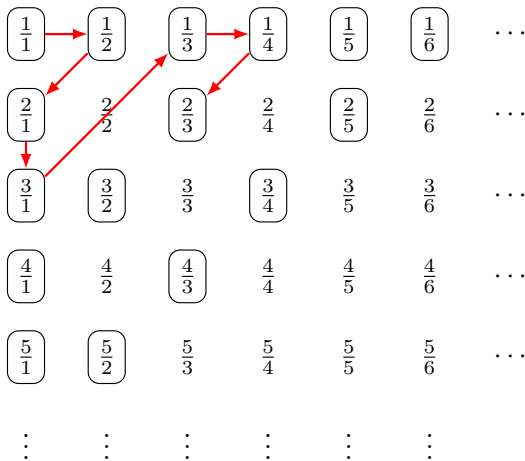
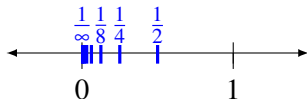
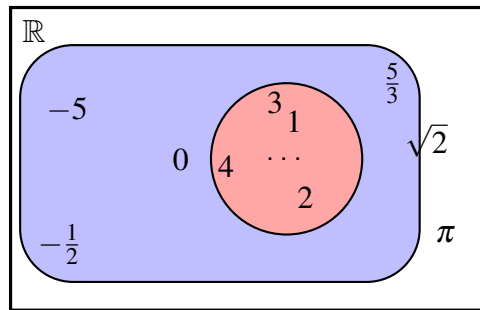
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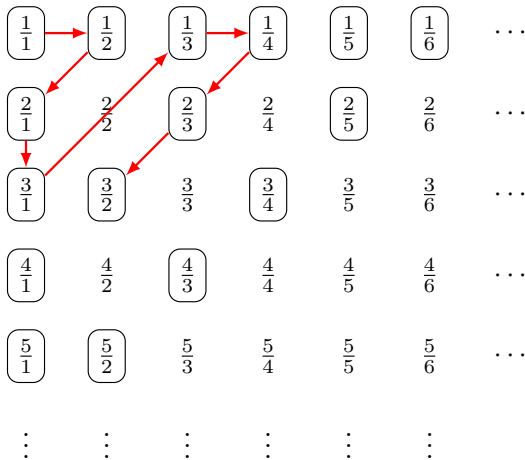
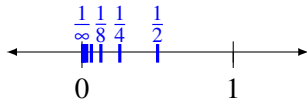
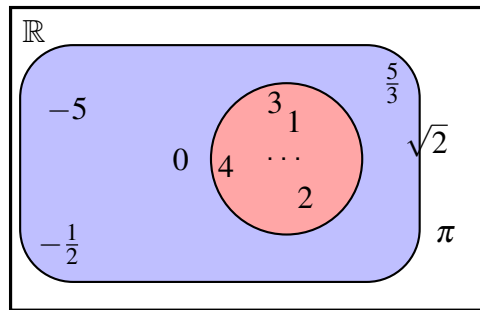
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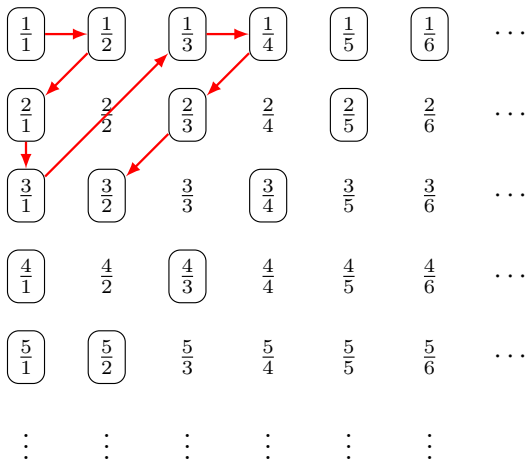
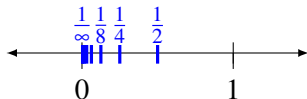
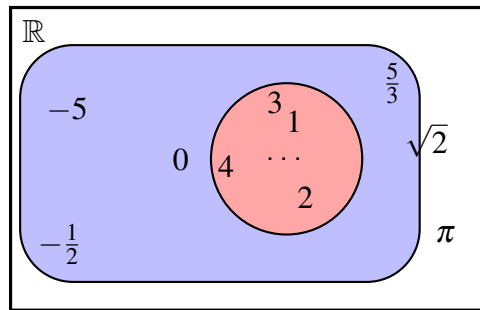
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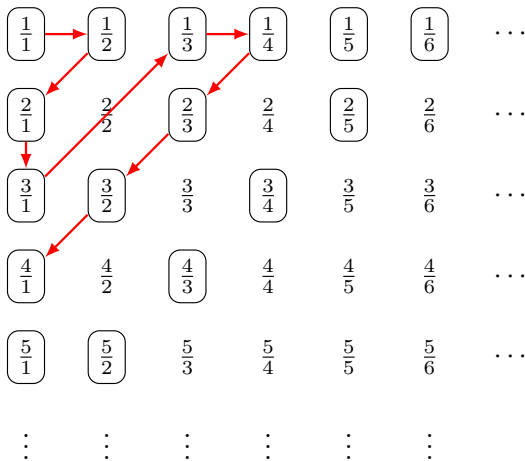
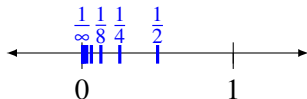
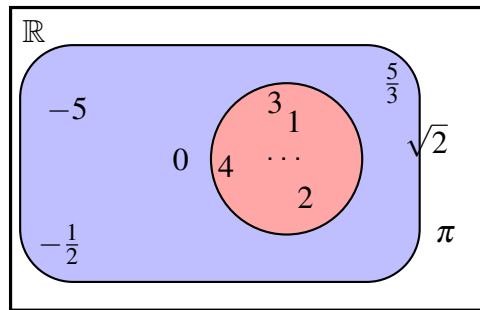
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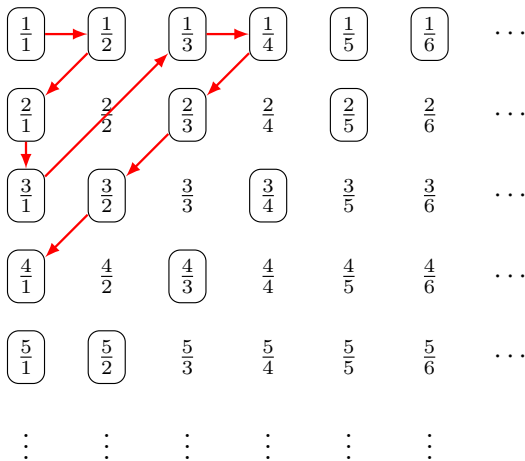
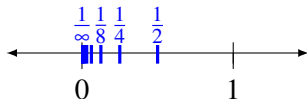
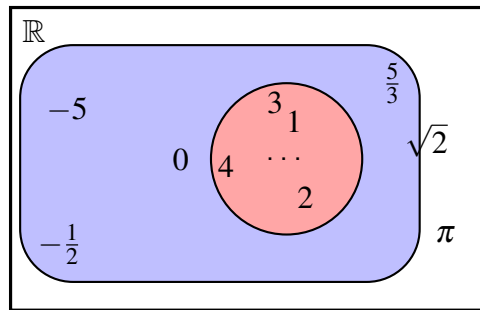
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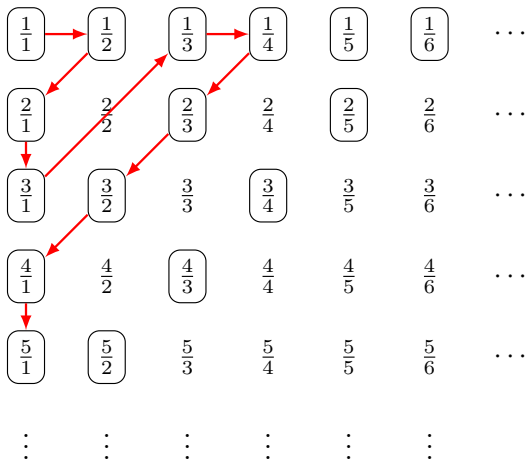
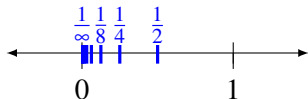
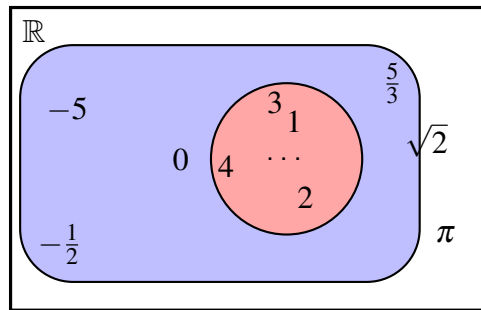
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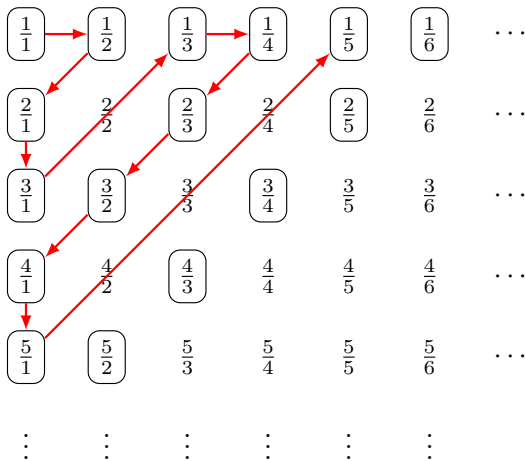
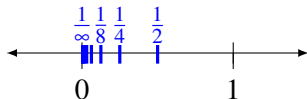
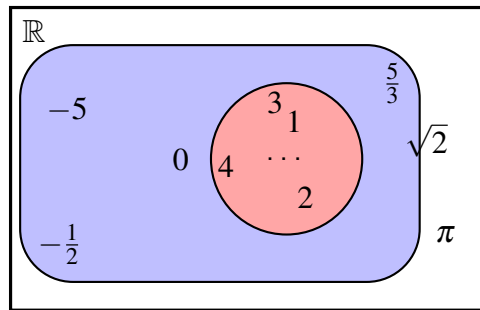
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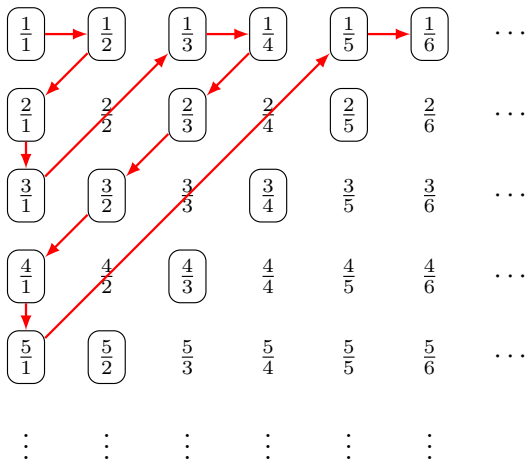
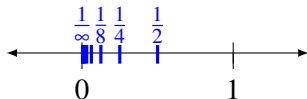
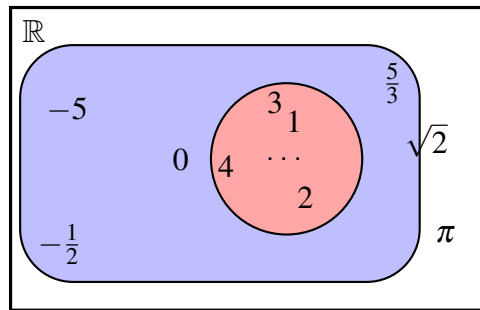
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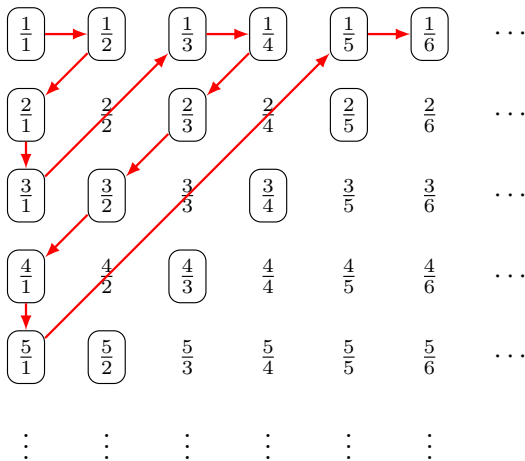
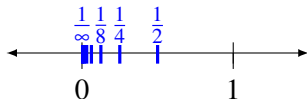
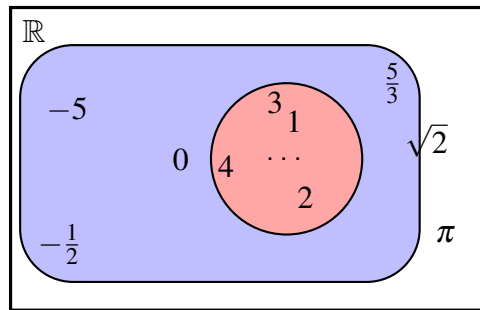
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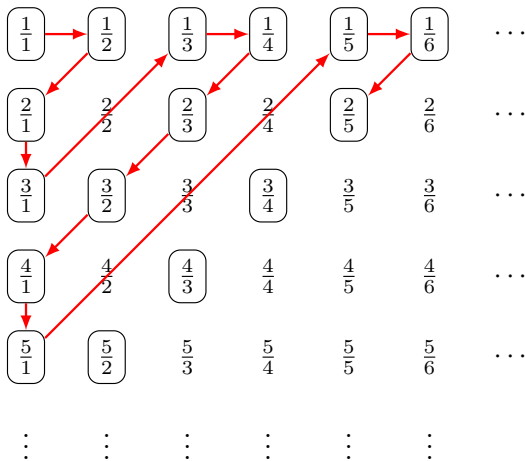
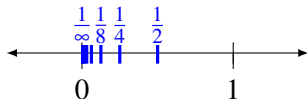
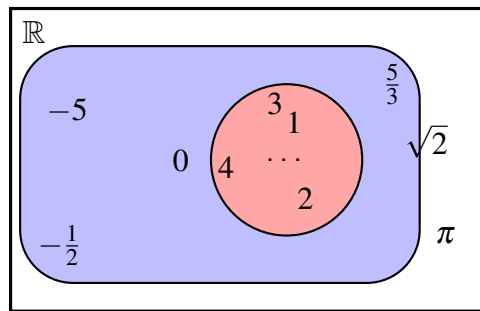
$$\mathbb{Q} = \{\frac{1}{2}, \frac{3}{4}, \frac{5}{3}, -1, 4, 0, \dots\}$$



Are these two sets the same size?

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

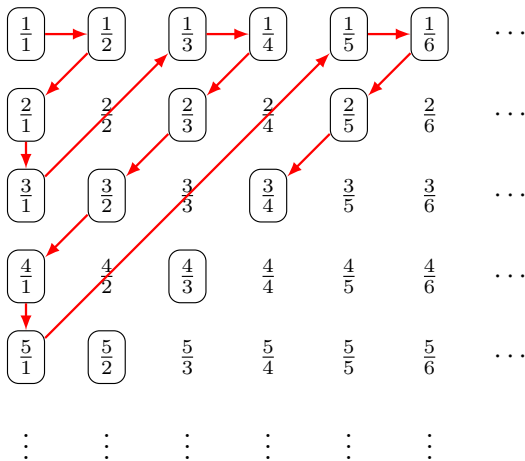
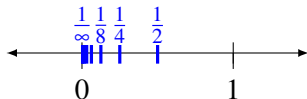
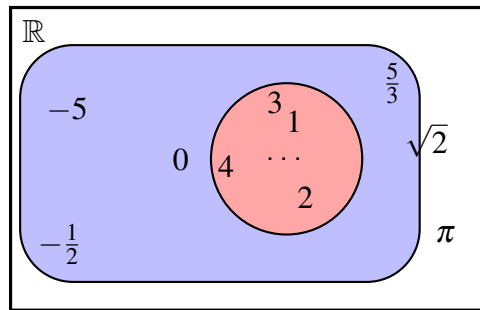
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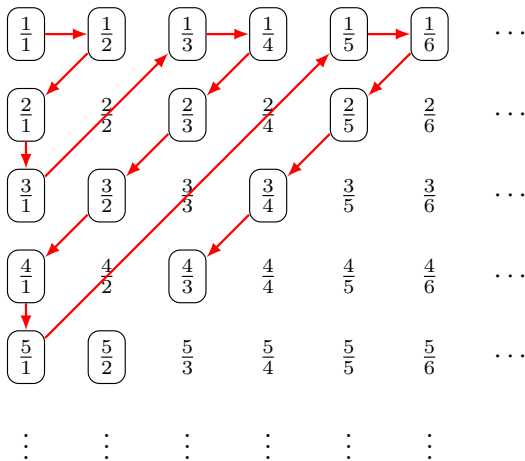
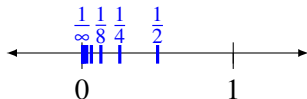
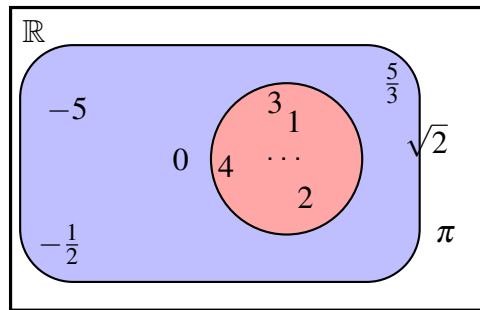
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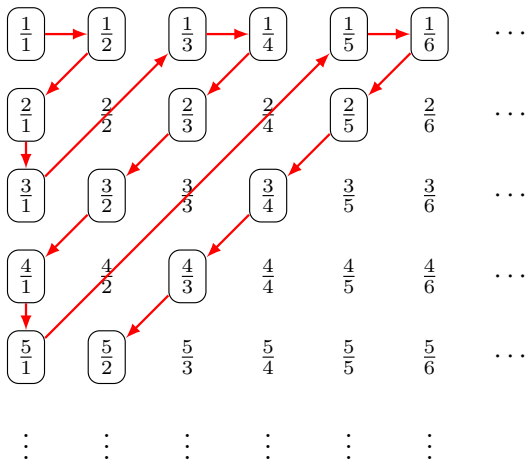
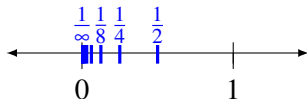
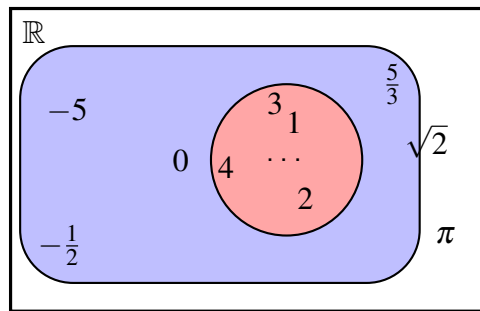
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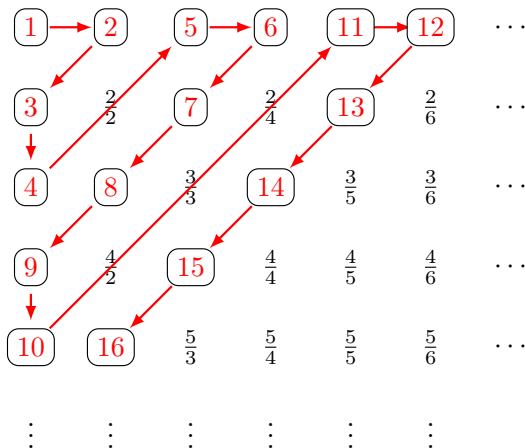
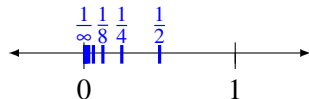
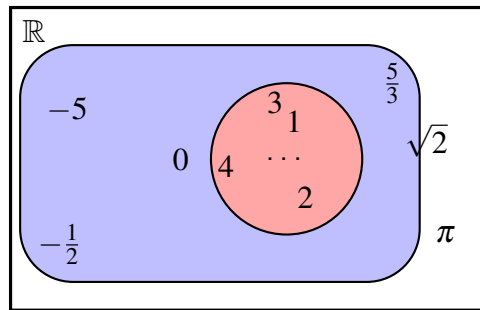
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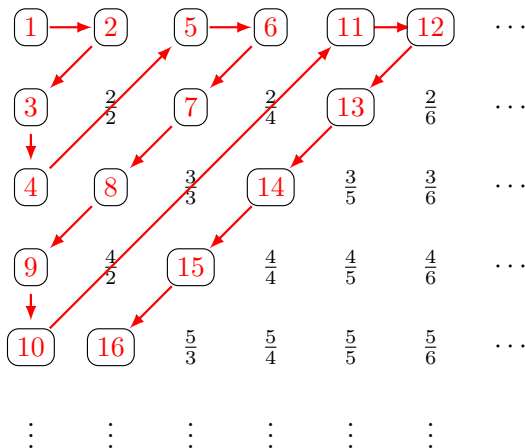
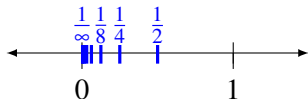
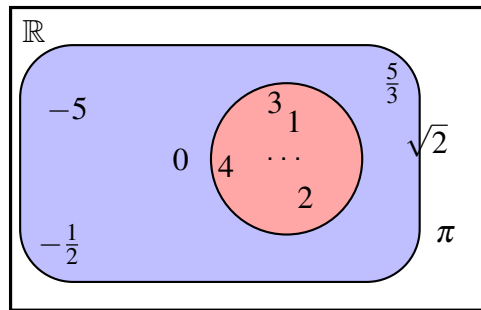
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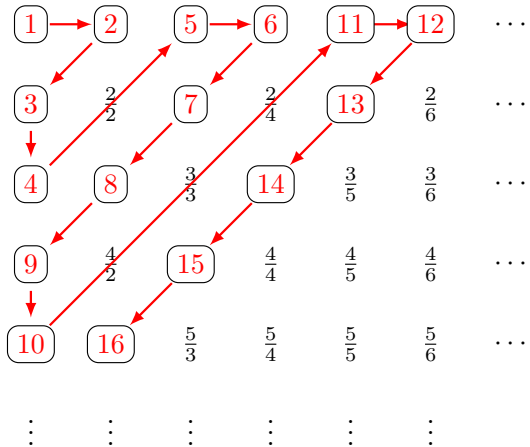
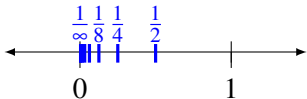
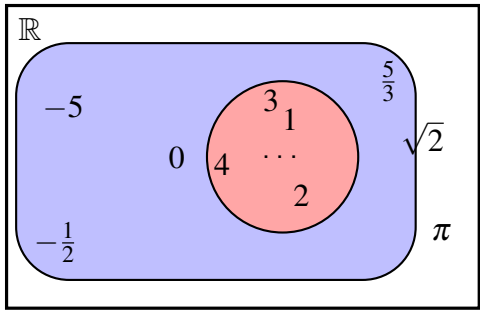


Are these two sets the same size?

Yes!

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

$$\mathbb{Q} = \{\frac{1}{2}, \frac{3}{4}, \frac{5}{3}, -1, 4, 0, \dots\}$$



These are example of "direct" proofs.

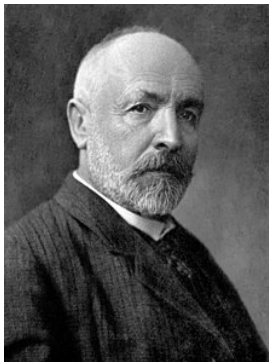
Are all infinite sets the same size?

Are all infinite sets the same size?

$$|\infty| = |\infty|?$$

Are all infinite sets the same size?

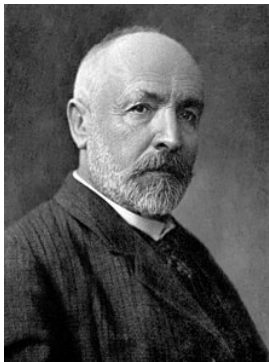
$$|\infty| = |\infty|?$$



Georg Cantor (1845-1918)

Are all infinite sets the same size?

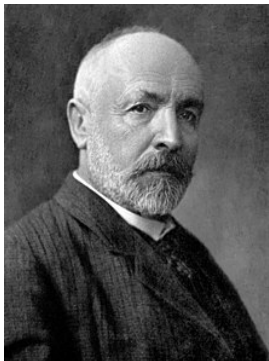
$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

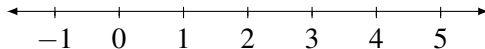
Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



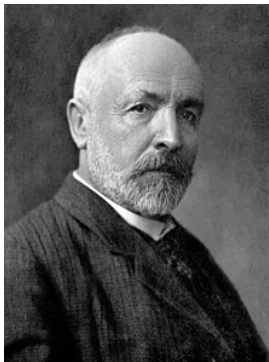
Georg Cantor (1845-1918)

What are the *Real* numbers ($\mathbb{R}$)?



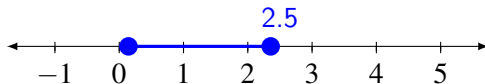
Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



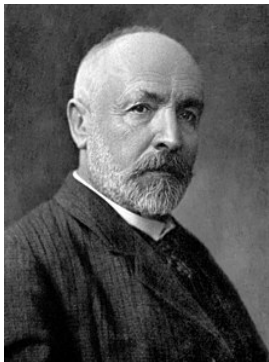
Georg Cantor (1845-1918)

What are the *Real* numbers ($\mathbb{R}$)?



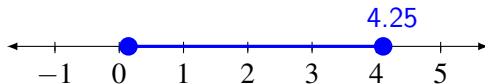
Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



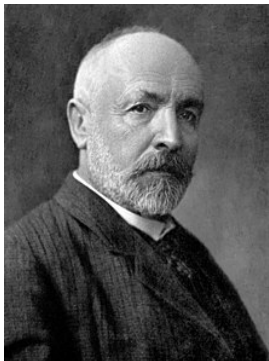
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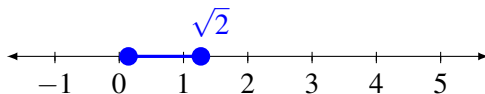
Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



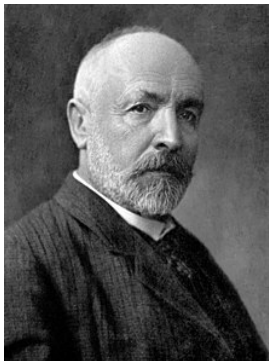
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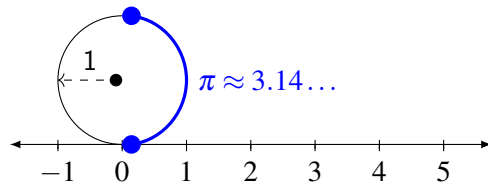
Are all infinite sets the same size?

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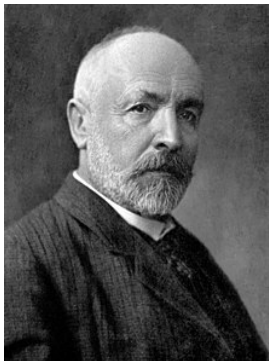
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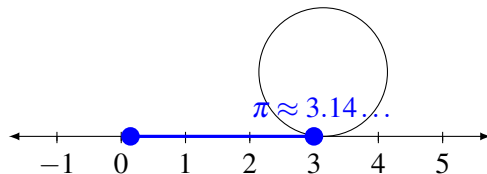
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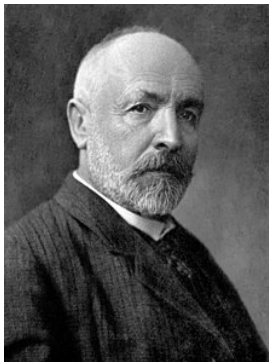
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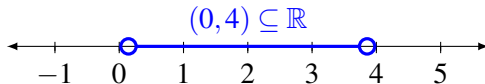


Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$

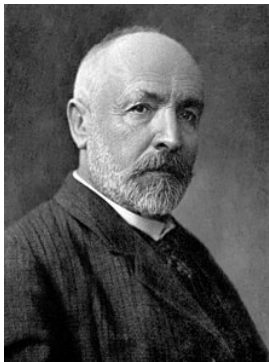


Georg Cantor (1845-1918)

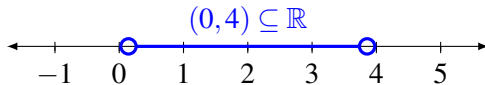


Are all infinite sets the same size?

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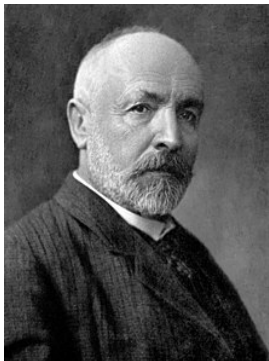


Georg Cantor (1845-1918)



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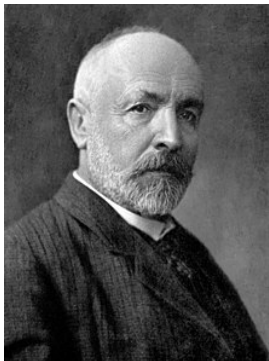


Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



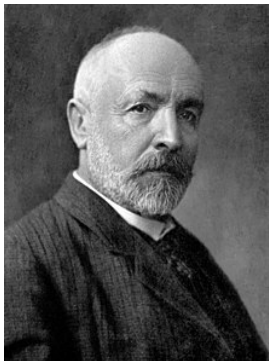
Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

$$1 \rightarrow 3.1415\dots$$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

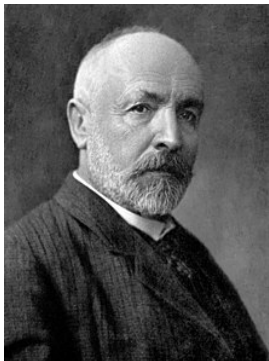
Assume this set is *countable*, then you could write out a list, like this:

$1 \rightarrow 3.1415\dots$

$2 \rightarrow 2.5000\dots$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

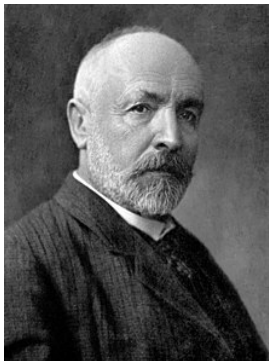
$1 \rightarrow 3.1415\dots$

$2 \rightarrow 2.5000\dots$

$3 \rightarrow 0.1250\dots$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



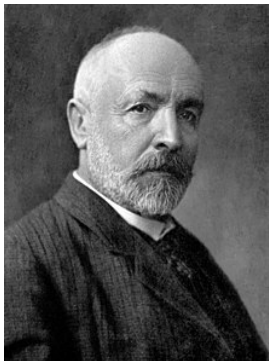
Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

$$\begin{array}{l} 1 \rightarrow 3.1415\dots \\ 2 \rightarrow 2.5000\dots \\ 3 \rightarrow 0.1250\dots \\ 4 \rightarrow 3.9999\dots \\ 5 \rightarrow 1.2345\dots \\ \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \end{array}$$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

$1 \rightarrow 3.1415\dots$

$2 \rightarrow 2.5000\dots$

$3 \rightarrow 0.1250\dots$

$4 \rightarrow 3.9999\dots$

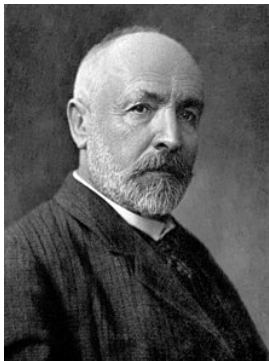
$5 \rightarrow 1.2345\dots$

$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$

d

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

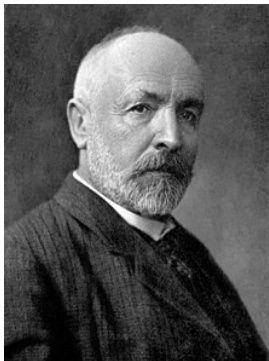
Assume this set is *countable*, then you could write out a list, like this:

1 $\rightarrow$ 3.1415...
2 $\rightarrow$ 2.5000...
3 $\rightarrow$ 0.1250...
4 $\rightarrow$ 3.9999...
5 $\rightarrow$ 1.2345...
 $\vdots$ $\vdots$ $\vdots$ $\vdots$ $\vdots$ $\vdots$

d

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

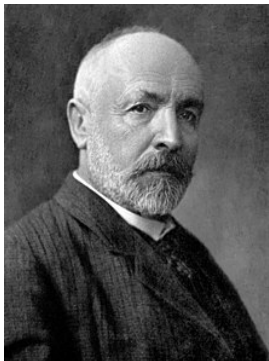
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2 $\rightarrow$ 2.5000...
3 $\rightarrow$ 0.1250...
4 $\rightarrow$ 3.9999...
5 $\rightarrow$ 1.2345...
 $\vdots$ $\vdots$ $\vdots$ $\vdots$ $\vdots$ $\vdots$

$$d = 2.$$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

$$1 \rightarrow 3.1415\dots$$

$$2 \rightarrow 2.\mathbf{5}000\dots$$

$$3 \rightarrow 0.1250\dots$$

$$4 \rightarrow 3.9999\dots$$

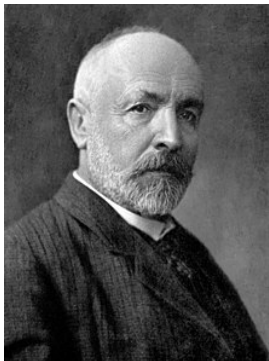
$$5 \rightarrow 1.2345\dots$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

$$d = 2.$$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

$$1 \rightarrow 3.1415\dots$$

$$2 \rightarrow 2.\mathbf{5}000\dots$$

$$3 \rightarrow 0.1250\dots$$

$$4 \rightarrow 3.9999\dots$$

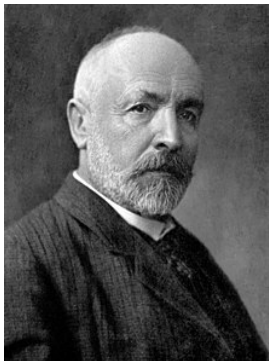
$$5 \rightarrow 1.2345\dots$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

$$d = 2.4$$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

$$1 \rightarrow 3.1415\dots$$

$$2 \rightarrow 2.5000\dots$$

$$3 \rightarrow 0.1250\dots$$

$$4 \rightarrow 3.9999\dots$$

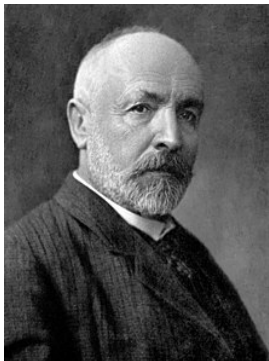
$$5 \rightarrow 1.2345\dots$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

$$d = 2.4$$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

$$1 \rightarrow 3.1415\dots$$

$$2 \rightarrow 2.5000\dots$$

$$3 \rightarrow 0.1250\dots$$

$$4 \rightarrow 3.9999\dots$$

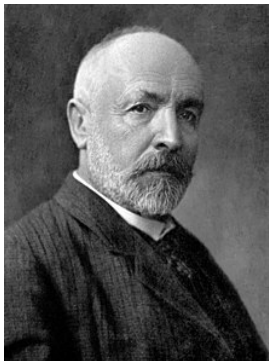
$$5 \rightarrow 1.2345\dots$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

$$d = 2.41$$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

$$1 \rightarrow 3.1415\dots$$

$$2 \rightarrow 2.5000\dots$$

$$3 \rightarrow 0.1250\dots$$

$$4 \rightarrow 3.99\mathbf{9}9\dots$$

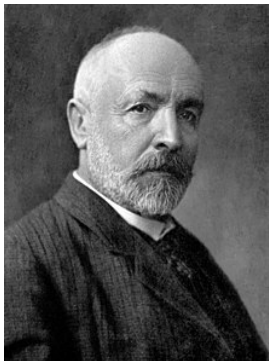
$$5 \rightarrow 1.2345\dots$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

$$d = 2.41$$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

$$1 \rightarrow 3.1415\dots$$

$$2 \rightarrow 2.5000\dots$$

$$3 \rightarrow 0.1250\dots$$

$$4 \rightarrow 3.99\mathbf{9}9\dots$$

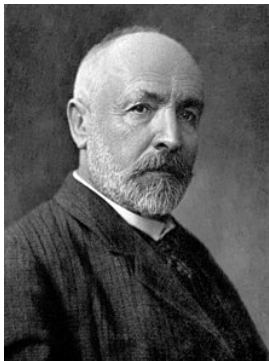
$$5 \rightarrow 1.2345\dots$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

$$d = 2.418$$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

$$1 \rightarrow 3.1415\dots$$

$$2 \rightarrow 2.5000\dots$$

$$3 \rightarrow 0.1250\dots$$

$$4 \rightarrow 3.9999\dots$$

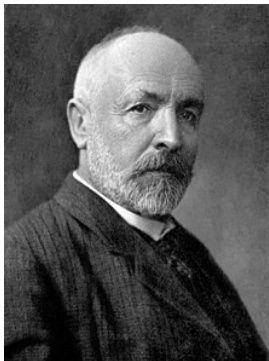
$$5 \rightarrow 1.234\mathbf{5}\dots$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

$$d = 2.418$$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

$$1 \rightarrow 3.1415\dots$$

$$2 \rightarrow 2.5000\dots$$

$$3 \rightarrow 0.1250\dots$$

$$4 \rightarrow 3.9999\dots$$

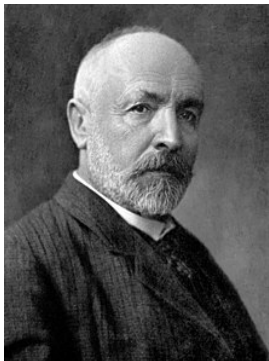
$$5 \rightarrow 1.234\mathbf{5}\dots$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$

$$d = 2.4184$$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

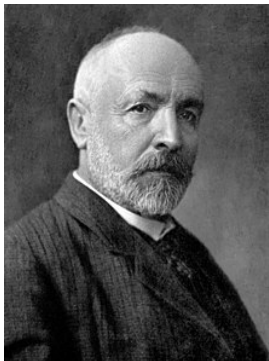
Assume this set is *countable*, then you could write out a list, like this:

1 $\rightarrow$ 3.1415...
2 $\rightarrow$ 2.5000...
3 $\rightarrow$ 0.1250...
4 $\rightarrow$ 3.9999...
5 $\rightarrow$ 1.2345...
 $\vdots$ $\vdots$ $\vdots$ $\vdots$ $\vdots$ $\vdots$

$$d = 2.4184$$

Are all infinite sets the same size?

$$|\{1, 2, 3, 4, \dots\}| = |\mathbb{R}|?$$



Georg Cantor (1845-1918)

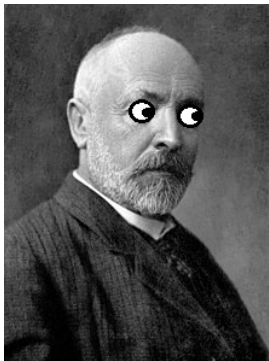
Assume this set is *countable*, then you could write out a list, like this:

1 $\rightarrow$ 3.1415...
2 $\rightarrow$ 2.5000...
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4 $\rightarrow$ 3.9999...
5 $\rightarrow$ 1.2345...
 $\vdots$ $\vdots$ $\vdots$ $\vdots$ $\vdots$ $\vdots$

$d = 2.4184\dots$

Are all infinite sets the same size?

No -different infinities!



Georg Cantor (1845-1918)

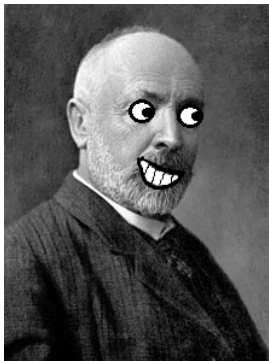
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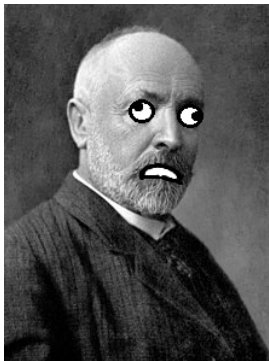
Georg Cantor (1845-1918)

Assume this set is *countable*, then you could write out a list, like this:

$$\begin{array}{l} 1 \rightarrow 3.1415\dots \\ 2 \rightarrow 2.5000\dots \\ 3 \rightarrow 0.1250\dots \\ 4 \rightarrow 3.9999\dots \\ 5 \rightarrow 1.2345\dots \\ \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \end{array}$$
$$d = 2.4184\dots$$

Are all infinite sets the same size?

No -different infinities!



Georg Cantor (1845-1918)

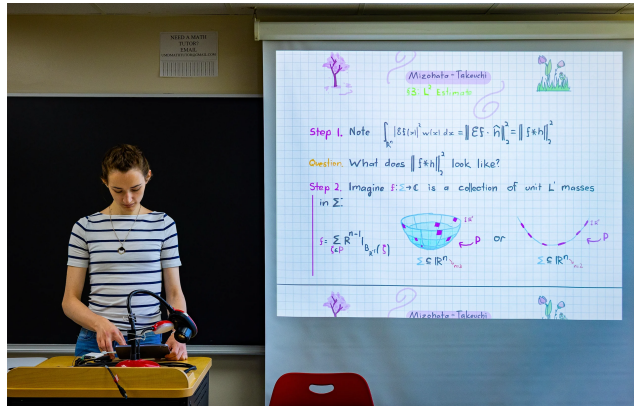
Assume this set is *countable*, then you could write out a list, like this:

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5 $\rightarrow$ 1.2345...
 $\vdots$ $\vdots$ $\vdots$ $\vdots$ $\vdots$ $\vdots$

$d = 2.4184...$

Proof by *contradiction*

Proof by contradiction



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Mizohata-Takeuchi
§3. L^2 Estimate

Step 1. Note $\int_{\mathbb{R}^n} |\delta f(x)|^2 w(x) dx = \|\mathcal{E}f \cdot \hat{h}\|_2^2 = \|f * h\|_2^2$

Question. What does $\|f * h\|_2^2$ look like?

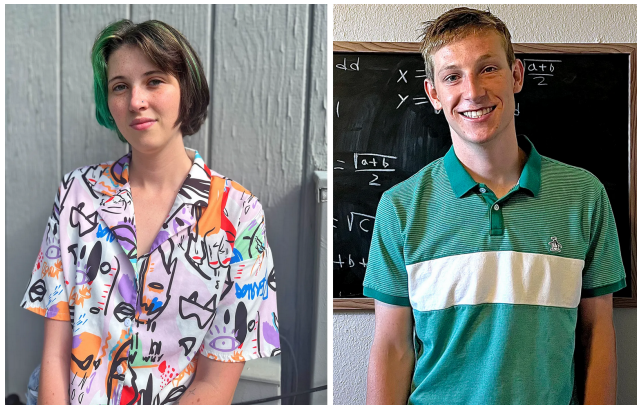
Step 2. Imagine $f: \mathbb{Z} \rightarrow \mathbb{C}$ is a collection of unit L^1 masses in $\mathbb{Z}$.

$f = \sum_{x \in \mathbb{Z}} \mathbb{R}^{n-1} |B_{x,1}(f)|$ or $\sum_{x \in \mathbb{R}^n} \mathbb{R}^{n-1} |B_{x,1}(f)|$

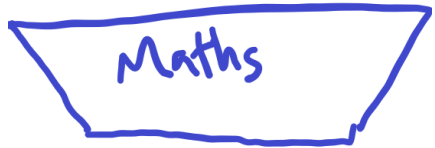
Mizohata-Takeuchi

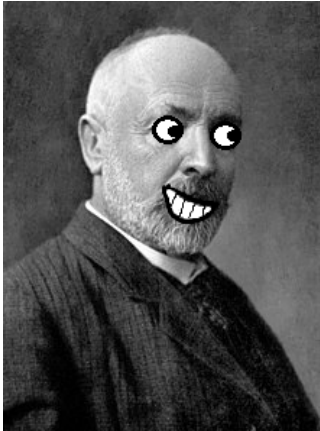
Mizohata-Takeuchi conjecture
Hannah Cairo

Proof by *contradiction*

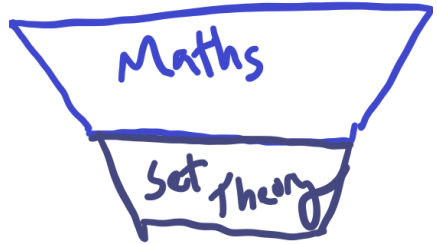


local-global conjecture
Summer Haag and Clyde Kertzer

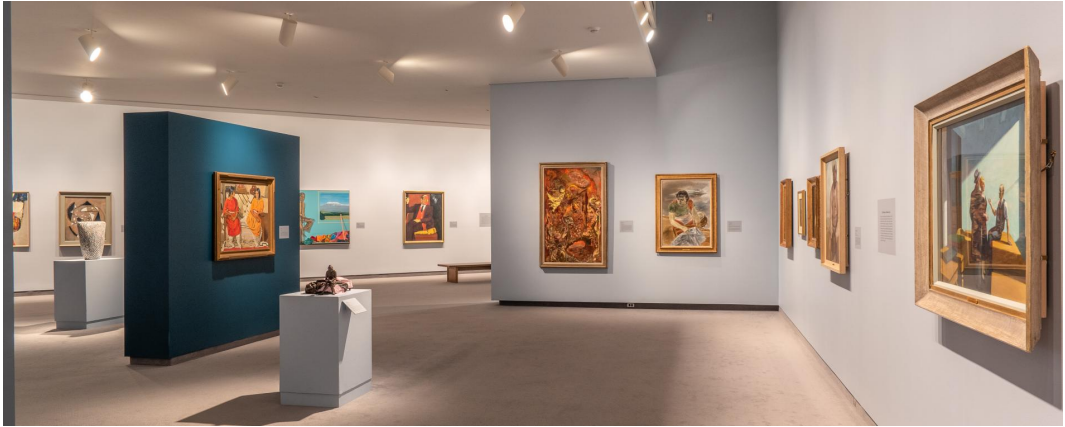




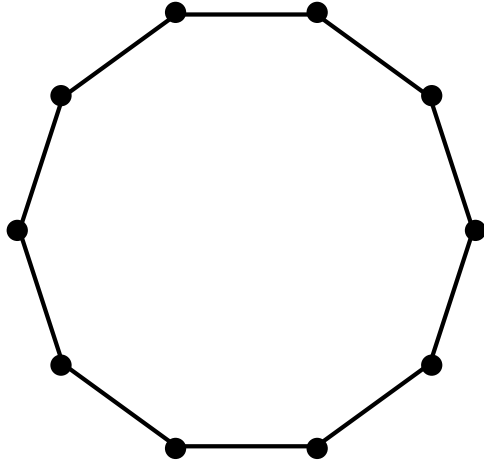
Georg Cantor (1845-1918)



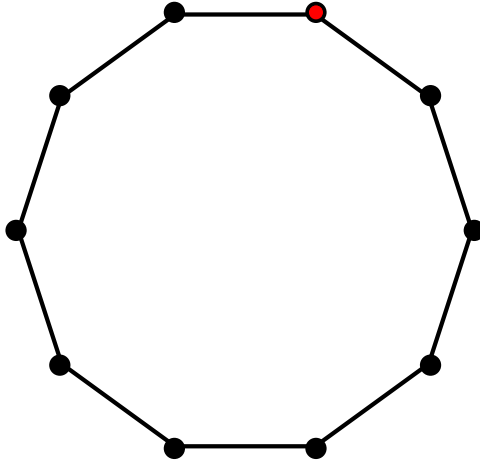
The Art Gallery Problem



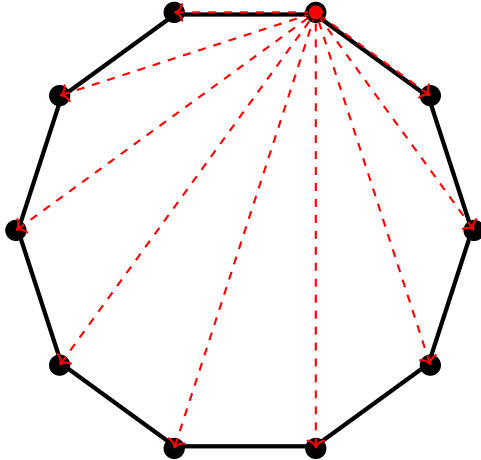
How many guards to guard the gallery? Easy case



How many guards to guard the gallery? Easy case



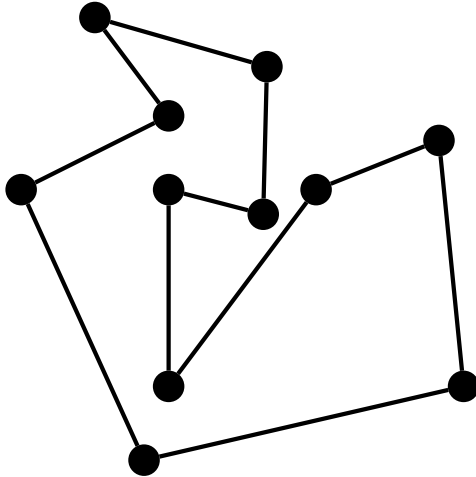
How many guards to guard the gallery? Easy case



Most guards needed to guard the gallery? fun case



Most guards needed to guard the gallery? fun case



Most guards needed to guard the gallery? fun case

Original proof by V. Chvatal (1975)

A Combinatorial Theorem in Plane Geometry

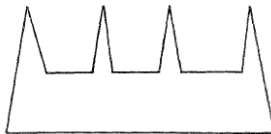
V. CHVÁTAL

*Centre de Recherches Mathématiques,
Université de Montréal, Montréal 101, Québec, Canada*

Communicated by W. T. Tutte

Received March 15, 1974

Let S be a subset of the Euclidean plane. We shall say that a subset A of S *dominates* S if for each $x \in S$ there is an $y \in A$ such that the entire segment xy lies within S . In a conversation, Professor Victor Klee asked for the smallest number $f(n)$ such that every set bounded by a simple closed n -gon is dominated by a set of $f(n)$ points. In a picturesque language, $f(n)$ can be interpreted as the minimum number of guards required to supervise any art gallery with n walls. Figure 1 shows that $f(12) \geq 4$; evidently, its pattern generalizes to yield $f(n) \geq \lceil n/3 \rceil$. We shall prove the reversed inequality in the setting of graph theory.



Most guards needed to guard the gallery? fun case

Original proof by V. Chvátal (1975)

40

V. CHVÁTAL

Proof. By induction on n . The cases $n = 3, 4, 5$ are trivial as each n -triangulation with $n \leq 5$ is a fan.

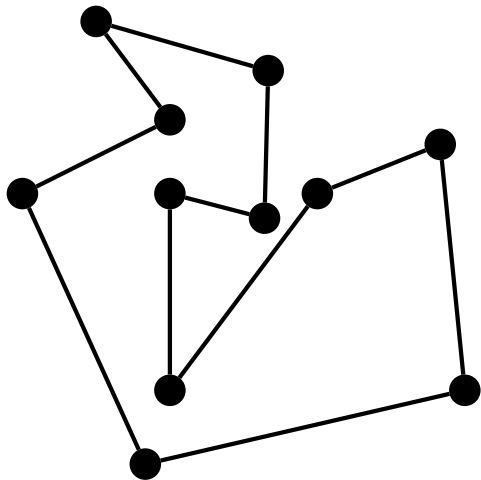
Now, let G be an n -triangulation with vertices $1, 2, \dots, n$ in their cyclic order. Let k be the smallest integer such that $k \geq 4$ and G has an edge $(j, j+k)$. First of all, let us note that $k \leq 6$. Indeed, let t be maximal with $1 \leq t \leq k-1$ and such that j is adjacent to $j+t$. To complete the triangle with sides $(j, j+t)$ and $(j, j+k)$, G must include the edge $(j+t, j+k)$. By the minimality of k , we must have $t \leq 3$, $k-t \leq 3$ and the desired inequality follows.

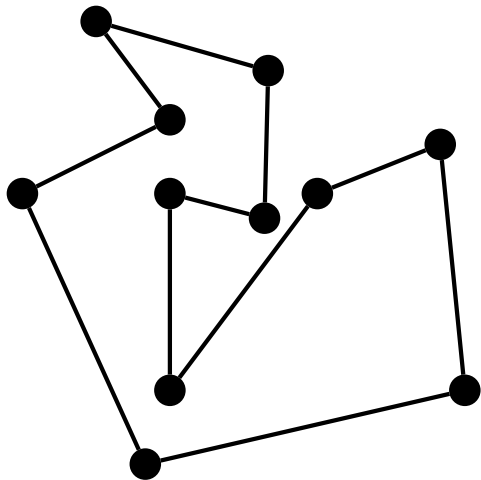
The edge $(j, j+k)$ cuts G into a $(k+1)$ -triangulation G_1 and an $(n-k+1)$ -triangulation G_2 . It is easy to verify that we have one of the following four cases (or perhaps a mirror image of (2) or (4)).

- (1) G_1 is a fan.
- (2) $k = 5$ and the inner edges of G_1 are $(j, j+2)$, $(j, j+3)$, $(j+3, j+5)$.
- (3) $k = 6$ and the inner edges of G_1 are $(j, j+2)$, $(j, j+3)$, $(j+3, j+6)$, $(j+4, j+6)$.
- (4) $k = 6$ and the inner edges of G_1 are $(j, j+3)$, $(j+1, j+3)$, $(j+3, j+6)$, $(j+4, j+6)$.

In Case 1, by the induction hypothesis, G_2 can be partitioned into m fans with $m \leq [(n-k+1)/3]$. Augmenting this partition by the fan G_1 , we obtain a partition of G into $m+1$ fans where $m+1 \leq [n/3]$.

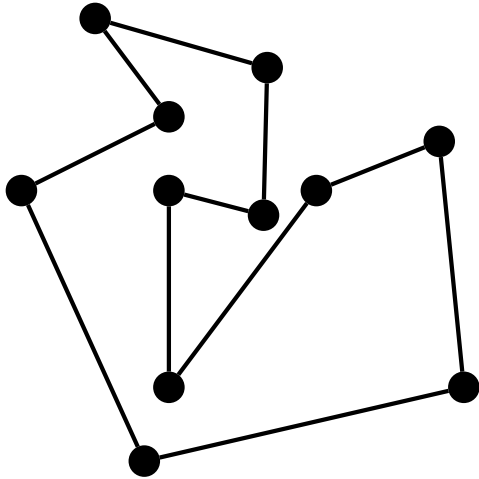
In Case 2, consider the $(n-3)$ -triangulation G_0 obtained from G_2





Claim:

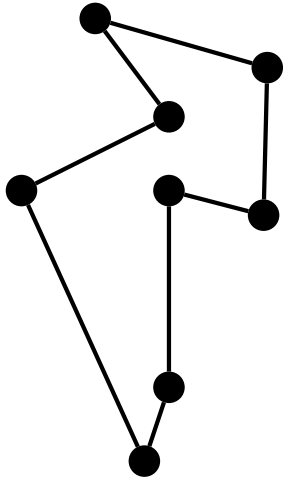
Any shape like this can be *triangulated*.



Claim:

Any shape like this can be *triangulated*.

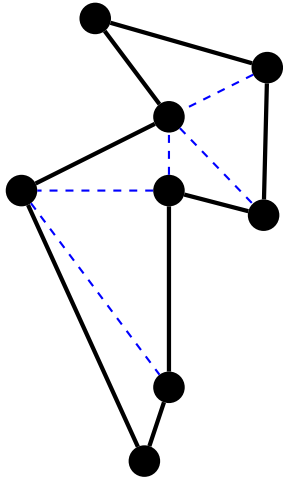
IF we can do it for N vertices.



Claim:

Any shape like this can be *triangulated*.

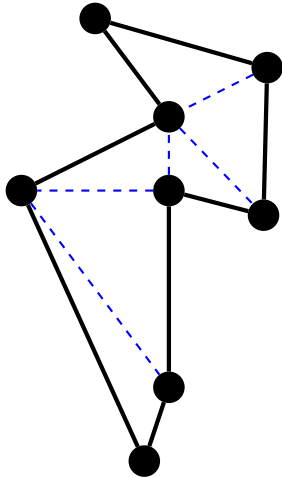
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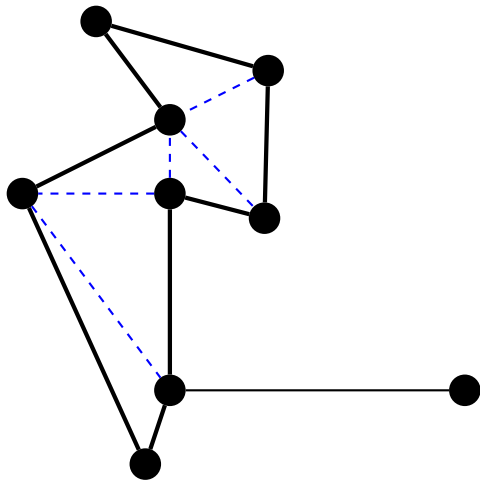


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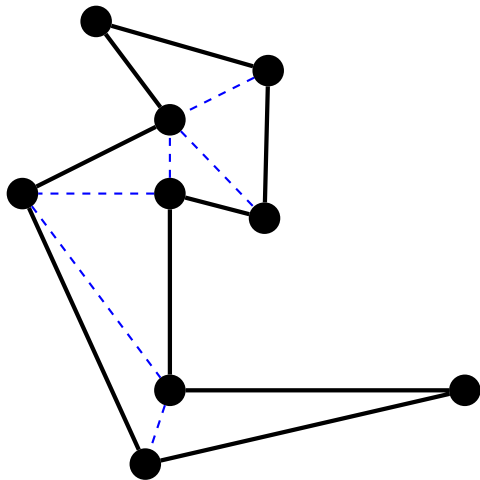




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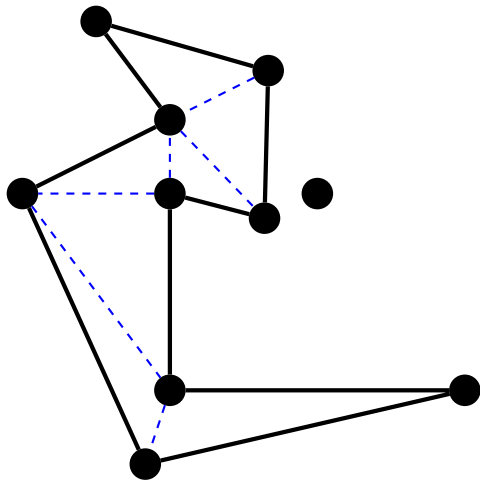


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Any shape like this can be *triangulated*.

IF we can do it for N vertices.

THEN we can do it for $N + 1$ vertices.

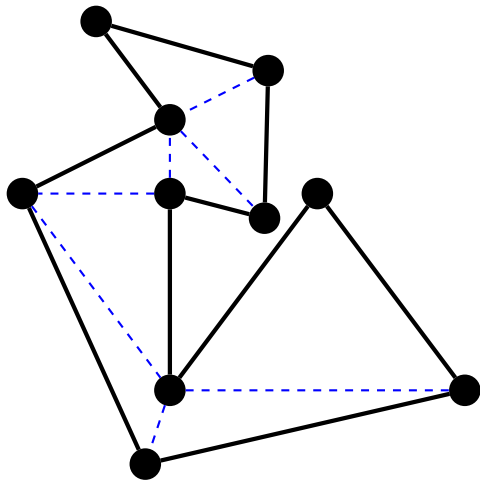


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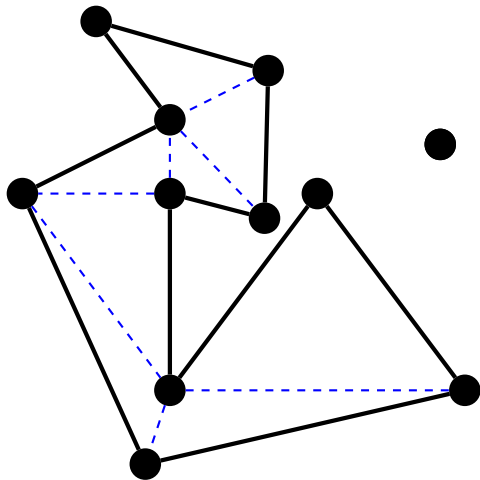


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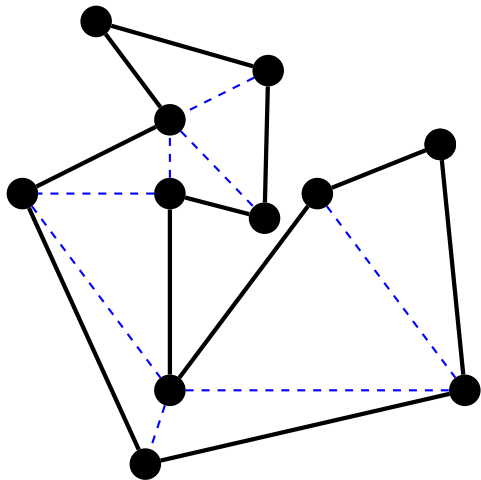


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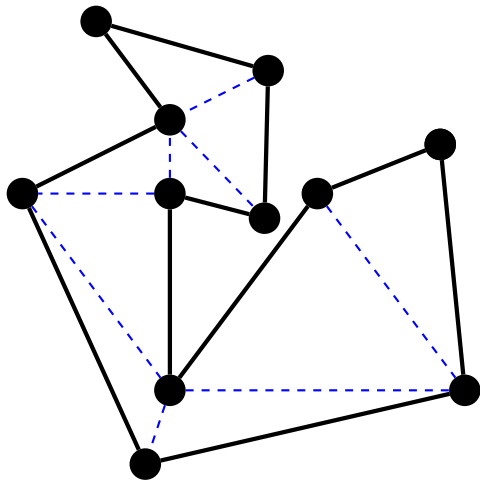


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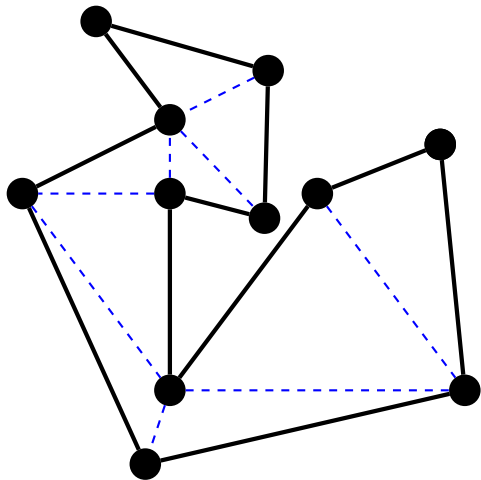
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So we just need a starting point...



Claim:

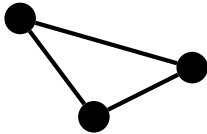
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$N = 3$



Claim:

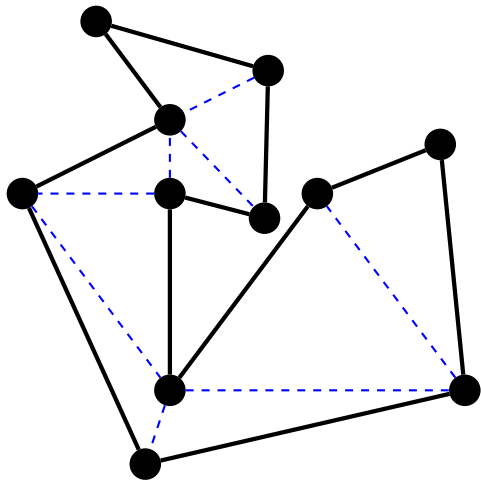
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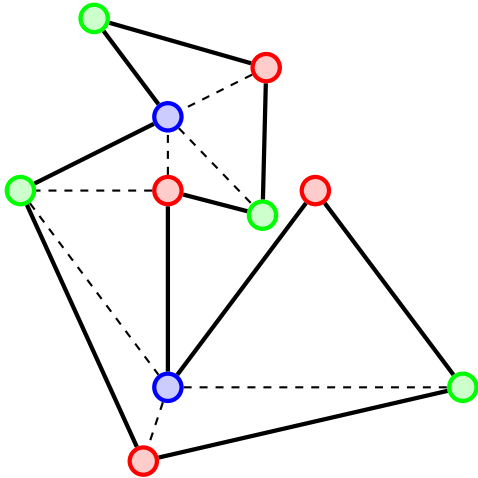
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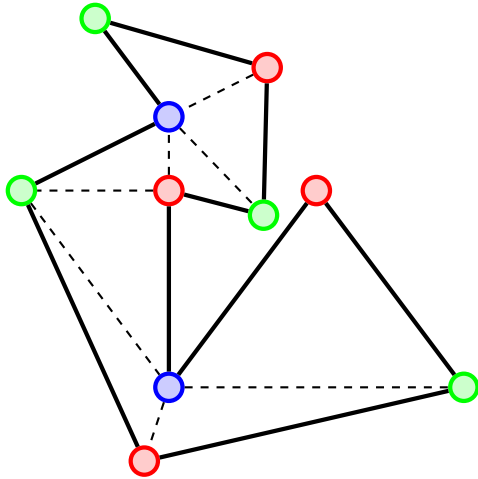
$$N = 3$$



Claim:

Any triangulated shape like can be *tri-colored*.

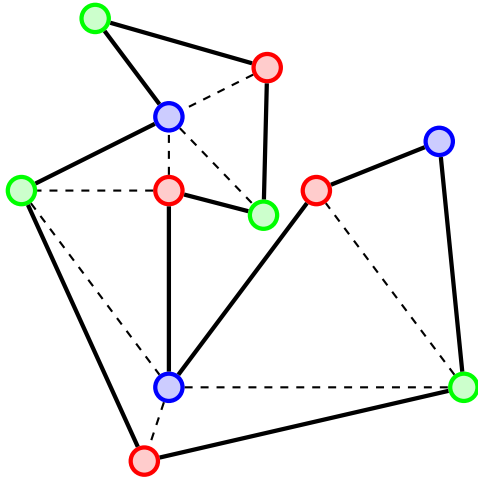




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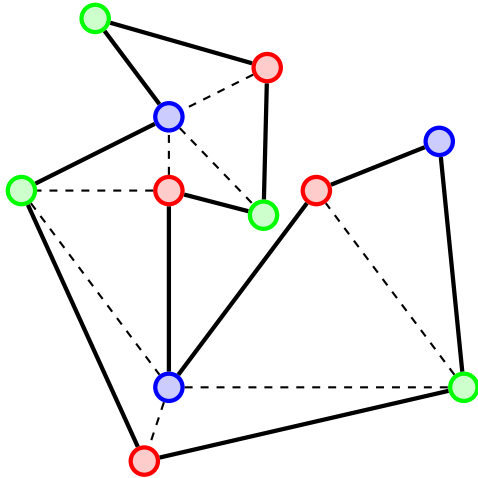


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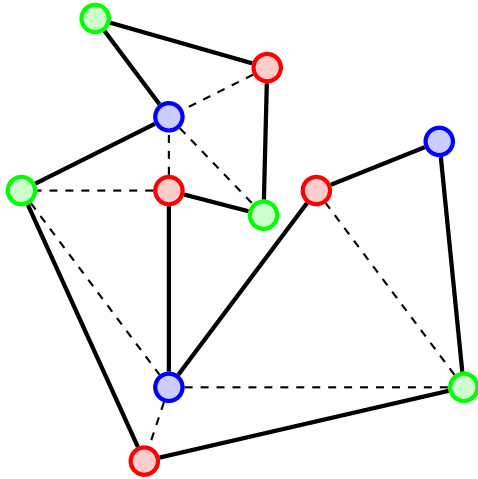
IF we can do it for N vertices.

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Theorem

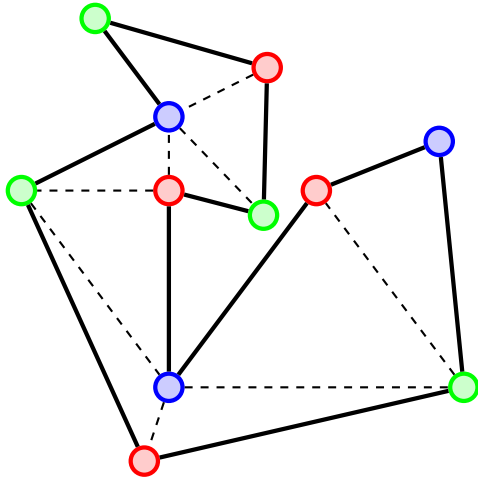
At most, $\lfloor \frac{N}{3} \rfloor$ guards are needed.



Theorem

At most, $\lfloor \frac{N}{3} \rfloor$ guards are needed.

$\lfloor \frac{N}{3} \rfloor$ is just $\frac{N}{3}$ rounded down.

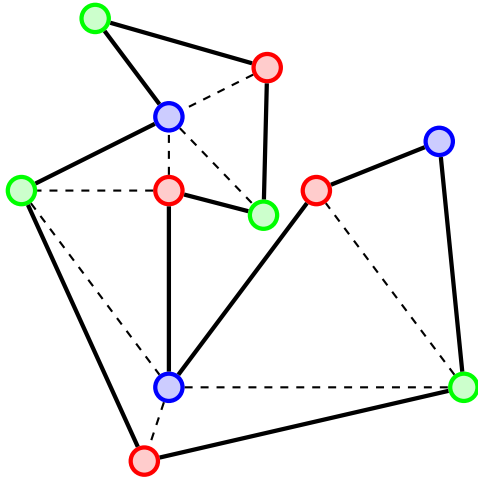


Theorem

At most, $\lfloor \frac{N}{3} \rfloor$ guards are needed.

$\lfloor \frac{N}{3} \rfloor$ is just $\frac{N}{3}$ rounded down.

Since there are N vertices, at least one color will occur $\lfloor \frac{N}{3} \rfloor$ times.



Theorem

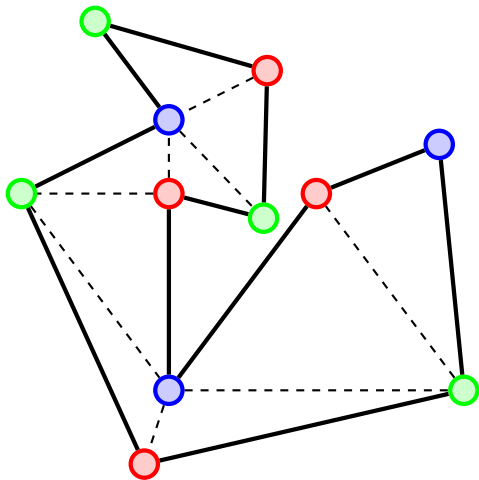
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Otherwise the colors would add up to more than N .

Post guards at the color!



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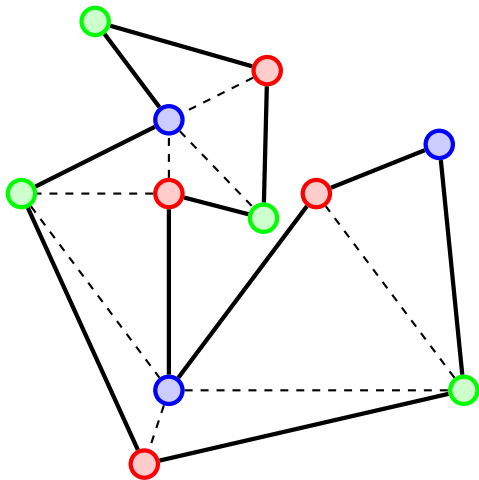
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$$\frac{11}{3} = 3.666\dots \text{ so 3 guards}$$

Proof by Induction

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Assume it's true for N .

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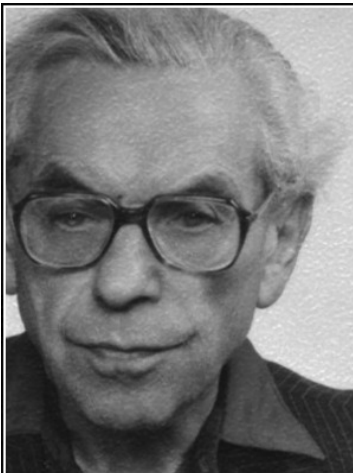
Show it's true for $N + 1$ (induction step).

Proof by Induction

Assume it's true for N .

Show it's true for $N + 1$ (induction step).

Show it's true for any N (base case).



God has the Big Book, the beautiful
proofs of mathematical theorems
are listed here.

— *Paul Erdős* —

AZ QUOTES